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Under pressure: how time constraints, task complexity, and AI reliability shape human-AI interaction

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ABSTRACT

This study investigates the impact of time pressure on human-AI collaboration in knowledge work environments. Drawing on dual-processing frameworks, specifically the heuristic-systematic model, we conducted an online experiment (n = 228) in which participants evaluated AI-generated responses across a series of tasks varying in complexity and correctness of AI responses. Participants subjected to a time-pressured treatment were compared to a control group with unlimited evaluation time. Our findings reveal that time pressure shifts cognitive processing from a systematic to a heuristic mode as expected, leading to diminished performance due to a reduced capacity to discriminate between correct and faulty AI responses. However, while both increased task complexity and AI advice faultiness independently impair human-AI-team performance, their interaction with time pressure suggests a more nuanced effect than predicted by traditional models. The results contribute to a refined understanding of cognitive dynamics in human-AI teaming and offer practical insights for designing AI systems and work environments that support effective decision-making under pressure.

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1. Introduction

The rapid advancement of artificial intelligence (AI) is fundamentally transforming the landscape of knowledge work, revolutionising how professionals across various domains approach problem-solving and decision-making (Chanda 2024; Dăniloaia and Turturean 2024; Gruetzemacher and Whittlestone 2022). As AI systems become increasingly sophisticated and ubiquitous, understanding the dynamics of human-AI collaboration has emerged as a critical area of research, with profound implications for organisational efficiency, innovation, and the future of work itself (Benbya, Pachidi, and Jarvenpaa 2021; Haefner et al. 2021; Jarrahi 2018). However, real-world scenarios often involve time constraints that can significantly impact human judgment and performance (Ranyard, Crozier, and Svenson 2002, 201), necessitating a deeper exploration of how these pressures influence the overall effectiveness of human-AI teams.

Emerging literature has documented the potential of human-AI collaboration, highlighting improvements in manageable workload, accuracy, processing speed, and the ability to handle complex information across various domains (Dell'Acqua et al. 2023; Lai, Kankanhalli, and Ong 2021). Concurrently, research has also examined the cognitive processes underlying human decision-making under time pressure, often drawing upon dual-processing frameworks of reasoning such as the *Heuristic-Systematic Model* (HSM; Chaiken 1980), the *Elaboration-Likelihood Model* (ELM; Cacioppo et al. 1986), or *fast* and *slow thinking* (Kahneman 2011) to explain shifts in information processing strategies.

According to these models, individuals utilise one of two cognitive pathways, the faster heuristic pathway or the more deliberate and effortful systematic pathway, based on available mental capacity, external cues, and boundary conditions, as well as individual factors such as motivation. Time pressure is usually considered a condition that shifts processing towards the heuristic pathway. Under time pressure, individuals tend to reduce information search and shift to less computationally demanding heuristics (e.g. acceleration, filtration, and strategy shifting; Spiliopoulos, Ortmann, and Zhang 2018). Initial evidence suggests that people are more likely to follow AI recommendations if ample decision time is available (Cao, Gomez, and Huang 2023). However, the extent to which time constraints influence reliance on AI-generated

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recommendations and overall performance in professional settings remains underexplored, presenting a critical blind spot in our ability to optimise human-AI teamwork.

In contrast to human-computer teams involving pre-LLM decision support and expert systems, today's AI systems warrant a fresh investigation into the effects at play. Distinguishing features include (1) instantaneous natural language interactions, resulting in an increasingly anthropomorphic perception by users (Nov, Singh, and Mann 2023), and (2) universal access by virtually anyone (almost) free of charge (in contrast to expensive and restricted expert systems available to exclusive groups only, as the name suggests). Moreover, these systems (3) are no longer restricted to specific domains but universally applicable across tasks and disciplines, and (4), are more powerful than systems of the past by an order of magnitude. They are also (5) capable of creating non-deterministic, *creative* outputs, while (6) providing context, explanations, and references to illustrate the underlying reasoning. Importantly, the broader societal impacts of these AI systems are contingent on how technological features, human goals, and contextual factors interact over time (Sabherwal and Grover 2024), emphasising the need to understand conditions like time pressure that shape human reliance on AI.

This study addresses this development by investigating the impact of time pressure on human-AI-team performance. By means of an online experiment, we investigated scenarios of collaboration between knowledge workers and AI systems. Participants were tasked to consider an AI's responses to various prompts across different domains and difficulties and to state their willingness to rely upon these responses.

In a nutshell, we find a negative effect of time pressure on human-AI performance. Intriguingly, our results suggest that time pressure does not unequivocally alter the degree to which humans rely on AI-generated answers, challenging some of HSM's major assumptions about how external factors such as time pressure or task difficulty cause humans to revert to heuristic decision-making and employ cognitive shortcuts. These insights provide a more nuanced understanding of human-AI collaboration, extending existing knowledge by illuminating the specific conditions under which time pressure affects team performance and decision-making processes.

The remainder of this paper is structured as follows: Section 2 presents a review of relevant literature and theory and develops our hypotheses based on the Heuristic-Systematic Model and existing research on human-AI collaboration as well as time pressure. Section 3 details our experimental design and methodology. In Section 4, we present our results. Finally, Section 5 discusses the implications of our findings for both theory and practice, acknowledges limitations, and illustrates paths for future work.

2. Related work

2.1. Time pressure and decision-making

Time pressure has emerged as one of the most important stressors in modern knowledge work environments due to new technologies, the expectation of ever-faster response times, as well as the 'always-available' nature of communication technology (Ninaus et al. 2015). Across the EU, 11% of workers report always working under time pressure and another 25% report doing so often (Eurostat 2020). As organisations adopt new technologies and practices into their operations, time pressure on knowledge workers keeps mounting (Hurtienne, Stilijanow, and Junghanns 2014).

The influence of time pressure on decision-making has been recognised in the domain of Information Systems for decades (Hwang 1994) and it is known to fundamentally alter human decision-making. Specifically, time pressure induces a speed-accuracy tradeoff, as decision-makers typically sacrifice precision for expedience. It manifests in reduced information search behaviours and increased reliance on heuristics (Spiliopoulos, Ortmann, and Zhang 2018). Decision-makers under time constraints often rely on pre-existing mental heuristics rather than thoroughly evaluating all available information. While this behaviour may enhance decision-making consistency by reducing random responses, it generally causes performance to deteriorate due to the application of oversimplified decision rules (Rice and Trafimow 2012). Additionally, time pressure amplifies information preferences, leading individuals to selectively focus on certain risk-related cues while ignoring others.

While some research suggests that time pressure may push decision-makers towards more extreme choices and reduced hedging behaviours (Madan, Spetch, and Ludvig 2015), other studies indicate that individuals do not optimally adjust their strategies under time constraints. Rather, they rely on fixed heuristics, which can lead to increased decision errors and, in some cases, system failures (Kerstholt 1994). Moreover, decision-makers initially become more conservative in low-risk situations but may take greater risks in high-risk scenarios under time pressure (Dror, Basola, and Busemeyer 1999). This effect is especially pronounced in contexts involving potential losses, where time pressure has been shown to significantly increase risk-seeking behaviour (Young et al. 2012).

Time pressure also increases strategic ignorance, where decision-makers intentionally avoid information that could complicate their choices, reducing cognitive load and enabling quicker, though potentially suboptimal decisions (Jarke-Neuert and Lohse 2022).

Moreover, individual differences play a significant role in how time pressure impacts decision-making. Experienced professionals may cope better with time constraints due to their expertise and familiarity with the task at hand, while novices may be more susceptible to the adverse effects of time pressure.

Time pressure in different phases of decision-making also influences human-AI collaboration, with longer final decision time increasing the likelihood of following AI suggestions (Cao, Gomez, and Huang 2023). While AI can facilitate rapid decisions by providing immediate recommendations, suboptimal integration and design can increase cognitive load, thereby amplifying the negative effects of time constraints. Additionally, time pressure may influence users' trust in AI recommendations, potentially leading to over-reliance when rushed or under-utilisation due to insufficient evaluation time.

Recent technological advances, such as AI-driven decision support systems and automation tools, aim to alleviate time pressure by streamlining information processing and offering quick insights. However, these tools may also introduce new complexities and require careful integration into existing workflows. To mitigate these negative impacts, individuals often employ strategies like prioritising tasks, enhancing time management skills, and applying stress-reduction techniques to improve decision-making.

In summary, decisions become riskier, faster, and more heuristic under time pressure, with individuals adjusting their strategies based on task familiarity and risk context. Consequently, the impact of time pressure on human-AI collaboration is multifaceted, affecting both AI supervision and advice uptake strategies. The observation of strategic ignorance and selective information usage further suggests that decision-makers may not always use AI recommendations optimally when rushed (Jarke-Neuert and Lohse 2022). This dynamic is reflected in our first hypothesis in the following section.

2.2. Human-AI collaboration

GenAI is transforming knowledge work by enhancing productivity, creativity, innovation, and flexibility (Benbya, Strich, and Tamm 2024). As genAI becomes integrated into workflows, tasks increasingly shift from manual creation based on individual skills to roles involving prompting, supervising, and assessing AI outputs (Palani and Ramos 2024). This AI augmentation of workers enables the production of high-quality outputs without necessarily possessing the skills required for manual creation (Ritala, Ruokonen, and Ramaul 2024). This also promotes machines from mere tools to agents who communicate, participate in task coordination, and possess a degree of agency, opening up a vast novel research agenda on the risks and benefits of '*machines as teammates*' (Seeber et al. 2020, 1). In contrast to an outright automation of tasks and decisions via AI systems, many current use cases of AI are structured as human-in-the-loop designs, either due to the standalone AI systems' lack of capabilities or with the human as a '*last line of defense against AI-failures*' (Passi and Vorvoreanu 2022, 1). As such, it is the human's role to effectively oversee the AI, to accept valid AI suggestions while overriding erroneous outputs. In practice, this complementarity in human-AI-collaboration is not easily achieved, oftentimes leading to suboptimal results compared to the standalone performance of humans or AI systems respectively (Schöffer et al. 2025). A growing body of research has further complicated this picture by highlighting that humans exhibit a significant algorithmic appreciation bias, whereby they tend to inherently trust algorithmic advice, even when such trust may not be warranted based on the actual quality or reliability of the AI's outputs (Sachin and Schecter 2024).

Considering this transformation of the human factor in knowledge work, human judgment, and critical evaluation skills emerge as new essential skills for ensuring the quality of work products, as only effective supervision of genAI will ensure the quality of results and mitigation of harm (Solaiman et al. 2023; Weidinger et al. 2023; Weisz et al. 2023).

Depending on the context, harm from inadequate oversight may manifest in different ways: In 2012, a glitch in a trading algorithm in conjunction with inadequate fail-safe mechanisms and lacking human supervision caused a sustained USD ~450 million loss within a 45-minute time frame (Securities and Exchange Commission 2013); a similar glitch in 2010 already caused a temporary loss of USD 1 trillion in market capitalisation in the Dow Jones Industrial Average Index (Securities and Exchange Commission & Commodity Futures Trading Commission 2010). In healthcare, AI systems providing false-negative diagnostics or inadequate treatment recommendations can pose life-threatening risks to patients if these errors are not corrected by human supervisors (Macrae 2019). In the realm of autonomous vehicles, self-driving cars have become disengaged or involved in crashes due to

misclassifying their surroundings (Leilabadi and Schmidt 2019). Although stakes in knowledge work environments usually tend to be less life-threatening[1], unintended AI behaviours – such as biases, hallucinations, or poor outputs resulting from flawed training data – can significantly impact the quality of work produced by genAI users.

The presence of time pressure in an environment with imperfect genAI outputs likely leads to a deterioration in the effectiveness of human supervision of the genAI. We therefore suggest that:

H_1 The presence of time pressure adversely affects human-AI-team performance.

2.3. Dual processing frameworks

The shift in cognition and processing of stimuli under the influence of stressors such as time pressure has long been conceptualised in dual processing frameworks. While nomenclature and conceptual details have evolved over time, frameworks such as the *Heuristic Systematic Model* (HSM; Chaiken 1980), the *Elaboration Likelihood Model* (ELM; Cacioppo et al. 1986), or *System 1 and System 2 Thinking* (Kahneman 2011) share the underlying concept of two separate pathways of human cognition that exist for automated and deliberate processing of information.

The HSM is considered the foundational theory in the domain and distinguishes *heuristic* (automatic, sub-conscious, low-effort) and *systematic* (deliberate, conscious, high-effort) modes of information processing (Chaiken 1980). Systematic processing is typically chosen for information of high personal relevance or general task importance, when information is unclear or complex, and when sufficient cognitive resources and time are available to the individual. In contrast, triggers for heuristic processing include familiarity with the information, the presence of heuristic cues that allow for mental shortcuts, low motivation for the task at hand, as well as a lack of cognitive resources or time.

Dual processing frameworks help understand how users process information, make decisions, and enact behaviours while engaging with digital systems, for instance, in digital healthcare, software development, and knowledge management (Watts 2015).

Given that genAI systems are fundamentally designed to augment human capabilities and enhance workforce productivity, it is crucial to understand how cognitive processing patterns influence the quality of human-AI interaction (Brynjolfsson and Mitchell 2017; Wilson and Daugherty 2018). As these systems become increasingly integrated into knowledge work and decision-making processes (Rahwan et al. 2019), ensuring that users can effectively calibrate trust and validate AI outputs under varying cognitive conditions becomes paramount for realising the technology's intended benefits while mitigating potential risks from errors, hallucinations, and inaccuracies (Hancock et al. 2011).

2.4. Task complexity

Task complexity refers to the component, coordinative, and dynamic complexity of a task (Wood 1986) as well as its baseline skill or tool requirements, which often necessitate the simultaneous processing of multiple pieces of information, the application of specialised skills, and the management of uncertainty (Bedny, Karwowski, and Bedny 2012). Such complexity places significant demands on cognitive capacity, prompting users to increasingly rely on genAI to manage and alleviate the cognitive load (Rodenburg et al. 2018). Task complexity – in the context of cognitive load theory – can be understood through the types of cognitive demands it imposes: (1) *intrinsic*, (2) *extraneous*, and (3) *germane load* (Sweller 1988, 2011). First, *intrinsic load* represents the inherent difficulty of the task itself, including the number of elements and their interactivity, which require substantial cognitive resources to manage. Second, *extraneous load* involves additional cognitive effort stemming from the way information is presented or structured; when genAI can alleviate this, users experience less unnecessary cognitive strain. Third, *germane load* is associated with the process of constructing new knowledge or skill development.

In high-complexity tasks, all three types of loads can increase simultaneously, creating a compounded demand on the user's cognitive resources. GenAI can serve as a vital support tool by reducing extraneous load and, in some cases, supporting germane load by aiding in complex information synthesis (Koć-Januchta et al. 2022). However, as intrinsic load rises with task complexity, users may increasingly rely on genAI for solutions, potentially reducing the opportunity for skill development and understanding. Thus, genAI's role in task complexity is twofold: it helps manage cognitive load but, if overly relied upon, may inadvertently stifle users' cognitive engagement and learning, as has been observed for other immersive automation technologies (Bueno-Vesga, Xu, and He 2021; Rodenburg et al. 2018).

The value of genAI in human-AI teams is therefore closely tied to task complexity; theoretically, the more complex the task, the greater the potential time and effort saved on the human side due to genAI assistance.

However, more complex tasks heighten the stakes of human oversight in genAI applications, as these tasks are challenging for humans to solve independently, difficult for genAI to handle accurately, and genAI is potentially impossible for human supervisors to evaluate objectively. Consequently, the utility of genAI in high-complexity contexts depends heavily on the accuracy and transparency of its outputs. When AI outputs are either erroneous or opaque, users may misjudge the assistance provided, resulting in over-reliance on inaccurate outputs or under-utilisation of beneficial guidance, both of which can compromise overall performance (Wienrich and Latoschik 2021).

In human-AI collaborations involving high task complexity, users are thus more likely to shift from systematic to heuristic processing. Under such conditions, the cognitive effort required for a thorough evaluation of AI output becomes prohibitively high, causing users to rely on heuristics based on superficial cues. This shift may lead users to base their judgments on interface aesthetics or past AI performance, cues that may not accurately reflect the system's current performance or reliability. Based on this, we assume that increased task complexity adversely affects human-AI-team performance (premise P_1). Moreover, we propose that:

> H_2 Time pressure amplifies the adverse effect of task complexity on human-AI-team performance.

2.5. AI accuracy and advice faultiness

Generative AI systems are inherently limited in accuracy due to challenges in data quality, reliance on probabilistic patterns rather than factual databases, and sometimes opaque reasoning capabilities, all of which hinder the model's ability to produce consistently accurate and contextually appropriate responses (Lenat and Marcus 2023; Tang et al. 2023). Additionally, the rapidly evolving nature of many knowledge domains further complicates accuracy, requiring regular updates to avoid outdated responses (Mulukuntla 2022). When faced with faulty advice, users may struggle to discern when the AI's outputs are reliable and when they are not.

This issue is bound to become more pronounced under conditions of time pressure or high task complexity, as heuristic processing is prioritised: Users unable to engage in systematic processing may instead rely on heuristic cues such as past interactions with the system or the transparency of its processes (Daronnat et al. 2021). They may trust AI outputs based on perceived system authority rather than accuracy when cognitive resources are strained (Shin, Zaid, and Ibahrine 2020). Even in the absence of strain, some users are easily deceived by untrustworthy AI systems which exaggerate their competence (Banovic et al. 2023). Additionally, cognitive biases introduced by heuristics increase the likelihood of systematic errors (Kleinmuntz 1985), further reinforcing user dependence on inaccurate AI outputs. Users may trust AI outputs even when they are inaccurate, simply because the system appears reliable based on superficial cues (Wienrich and Latoschik 2021) or may simply ignore cues for untrustworthiness (Krügel, Ostermaier, and Uhl 2022). We thus assume that increased AI advice faultiness adversely affects human-AI-team performance (P_2). Moreover, we suggest that

> H_3 Time pressure amplifies the adverse effect of AI advice faultiness on human-AI-team performance.

Our full research model is visualised in Figure 1 below:

3. Method

To evaluate our research model, we conducted an online experiment.

3.1. Treatment design

We employed a between-subjects design with (1) a control group (CTR) in which participants had unlimited

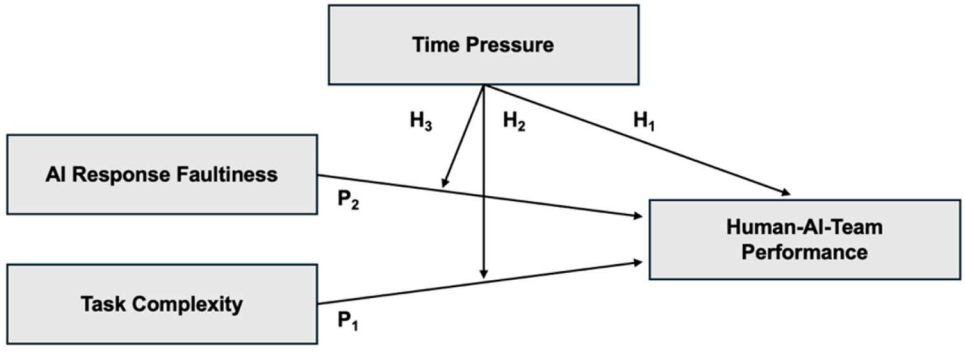


Figure 1. Research model.

time to complete the assigned tasks and (2) a treatment group (TP) in which participants faced time pressure. Based on the control group's recorded response times per task, the treatment group's time limits were set to the 30th percentile of observed response times per individual task. These values range between 6.9 and 28.1 s per task. Table 2 outlines the control group's median-, average-, and 30th-percentile response times per task.

3.2. Task and stimulus material

The experiment involved 36 tasks, presented in random order for each participant. In each task, participants were presented with a (written) prompt provided to ChatGPT as well as its response to the prompt. Participants' task was then to indicate their *willingness to rely* on this response on a scale from 0 (i.e. '*not trust response at all*') to 10 (i.e. '*fully trust the response*') as illustrated in Figure 2. We used a correct/ incorrect ratio of 3:1 (i.e. 75% correct, 25% incorrect).

We utilised ChatGPT's actual responses (at the time of conducting the experiment). All prompt-response pairs were created using the free version of ChatGPT 3.5 over the course several days by the author team. Each prompt was entered into a new session of ChatGPT and generally, only one response was generated per prompt. To obtain incorrect responses, some prompts had to be repeated in new sessions until an incorrect response was given by ChatGPT.[2]

Participants were briefed that, just as ChatGPT's actual responses in live interactions, some of the responses displayed in the experiment might be erroneous but were not provided with the actual ratio of correct/incorrect responses. Generally, we selected prompts for which there exist definitive, verifiable or falsifiable answers (i.e. objectively correct or incorrect; verifiable through logic, math or fact-checking). For the responses, where an objectively correct response might not unequivocally exist, (e.g. 'What's the most relevant academic paper on AI?'), we relied on internet research and discussions within the author team to determine whether an answer would be widely accepted as correct.

The tasks can be structured along nine categories of typical shortcomings in the capabilities of generative AI tools (see Table 1). These categories were adapted from an evolving collection of failures and weaknesses observed in ChatGPT's performance by various researchers (Borji 2023). Each participant was shown the full set of 36 tasks in random order while ensuring that faulty responses were not shown in direct consecutive order. The share of incorrect responses for each of the 36 experiment periods was in line with the expected share of 25% without any additional efforts for counterbalancing (see Appendix A for additional information and an exemplary participant sequence).

3.3. Task difficulty

As an important control variable, we assessed task difficulty in a separate auxiliary survey with 45 participants. Participants were asked to state how difficult they perceived to solve each prompt on a scale from 1 (very easy) to 10 (very difficult). While participants reported perceived difficulty levels across the entire scale, average difficulty levels per task ranged between 2.37 and 7.64. Average difficulty level per task is also provided in Table 2 below. We ensured similar average difficulties between tasks with correct (4.21) and incorrect (4.39) AI responses (two-sample t-test; $p = .773$).

3.4. Measurement instrument

For each task, we captured participants' stated *willingness to rely* on ChatGPT's response on an 11-point Likert scale from 0 (i.e. '*not trust response at all*') to 10 (i.e. '*fully trust the response*'). In a minority of cases (9.8%), participants in the time pressure treatment failed to respond within the time limit. We then set their response to a default value of 5, reflecting indecisiveness. Responses were incentivised such as that participants received an additional, variable payoff depending on how well their willingness to rely matched the response's correctness. For this *performance* value, we used the difference between a participant's *stated* willingness to rely and

Table 1. Prompt categories and rationale for inclusion in stimulus material.

Category	Rationale
Logic	Logical reasoning capabilities were lacking upon release of ChatGPT, even as responses appeared linguistically cohesive and plausible
Arithmetic	In its earlier releases, even simple arithmetic prompts would regularly yield false results
Factual	Prompts for factual information have objectively correct answers, yet were sometimes answered incorrectly by ChatGPT
Contemporary	ChatGPT used to have a knowledge cut-off date and wouldn't be able to include more recent events
Hallucination	Numerous cases of factually incorrect, completely fabricated responses to prompts were documented
Sequences	A sub-category of *Logic;* ChatGPT appeared inept at completing both semantic and arithmetic sequences
Brainteaser	A sub-category of *Logic*; complex logical reasoning utilising the Cognitive Reflection Test
Text-editing	Simple prompts to edit and assess texts would regularly yield outputs that do not adequately complete the prompted task
Code	ChatGPT would sometimes output non-functional, yet plausible-appearing code or pseudo-code

Table 2. Prompts, responses, and completion times.

Category	#	Task	Prompt	AI correct?	Response (shortened)		CTR response time (sec., n = 113)			Difficulty *(1–10, n = 45)*	
					Correct	ChatGPT	p_{30}	Median	Mean	Mean	Std. error
Logic	1	53_socks	A man has 53 socks in his drawer: 21 identical blue, 15 identical black, and 17 identical red. The lights are out, and he is completely in the dark. How many socks must he take out to make 100 percent certain he has at least one pair of black socks?	**No**	40 socks	16 socks	19.7	36.6	47.1	5.36	0.38
	2	two_fathers	Two fathers and two sons are in a car, yet there are only three people in the car. Is this situation possible?	Yes	Yes		17.6	22.8	30.1	2.64	0.39
	3	what_day	What day is tomorrow if the day before yesterday was the day after Sunday?	Yes	Thursday		24.0	34.6	44.1	3.02	0.34
	4	jack_anne	Jack is looking at Anne. Anne is looking at George. Jack is married, George is not, and we don't know if Anne is married. Is a married person looking at an unmarried person?	Yes	Yes		27.7	43.7	49.5	4.80	0.38
Arithmetic	5	seconds_in_day	How many seconds are in a day?	Yes	86,400		6.9	12.6	17.0	2.89	0.36
	6	calculation_1	What is the result of the following calculation: 334 * 7 + 335 = ?	Yes	2,673		13.8	20.9	25.7	3.53	0.45
	7	7_power_5	What's 7 raised to the power of 5?	Yes	16,807		11.1	16.8	22.5	4.24	0.50
	8	calculation_2	What is the result of the following calculation: 215*337?	**No**	72,455	72,555	8.5	13.0	18.9	3.31	0.43
Factual	9	mushroom	Which mushroom is the most poisonous in the world?	Yes	Death Cap (Amanita phalloides)		9.0	14.0	18.4	3.60	0.37
	10	lady_ermine	Who painted Lady with an Ermine?	Yes	Leonardo da Vinci		10.0	15.2	20.8	4.20	0.44
	11	state_neighbors	Which country has the most state neighbors?	**No**	Russia & China, both 14	China, 14	9.7	13.9	22.4	4.04	0.38
	12	hygrometer	What is measured with a hygrometer?	Yes	Humidity / moisture		7.8	11.7	18.4	3.78	0.44
Contemporary	13	vice_president	Who is the current Vice President of the United States and how many Vice Presidents have there been in total in U.S. history?	Yes	Kamala Harris / 49		11.8	16.8	24.6	3.78	0.37
	14	pdc_darts	Who won the most PDC darts world championship titles?	Yes	Phil Taylor		7.8	13.0	19.1	4.24	0.45
	15	argentina_world_cup	How many times did Argentina win the soccer World Cup?	**No**	3 times	2 times	10.3	14.5	20.9	3.13	0.39
	16	google_ceo	Who's the leader of Google's parent company Alphabet?	Yes	Sundar Pichai		11.6	17.8	23.3	3.58	0.40
Hallucination	17	paper_ai	What's the most relevant academic paper on AI? Please provide a citation in APA style.	Yes	'Deep learning' (LeCun et al., 2015)		11.1	17.8	24.4	7.42	0.37
	18	citation_apa	Please provide an example of an academic citation in APA style based on any paper from 1980 and include the document ID.	**No**	Correct citation / existing paper	Citation of hallucinated paper	9.9	14.4	31.9	6.73	0.41
	19	28th_amendment	What's the 28th amendment to the US constitution?	Yes	There is no 28th amendment to the US constitution		10.1	13.9	23.8	4.62	0.49
	20	antique_wonders	What are the ten antique wonders of the world?	Yes	There were only seven wonders of the world		12.8	18.2	26.1	3.91	0.43
Sequences	21	do_re_mi	What syllable is next in this sequence: Do-Re-Mi-Fa-So-La?	Yes	Ti		11.8	18.0	23.5	2.73	0.42
	22	zaybxc	Which letter comes next in the sequence: Z – A – Y – B – X – C –?	**No**	W	D	18.8	30.3	34.6	2.53	0.37
	23	number_sequence_1	Which number comes next in the sequence: 4 – 10 – 22 – 46 – 94 –?	Yes	190		14.1	23.6	40.9	3.16	0.38
	24	number_sequence_2	Find the next number in the series: 21 – 32 – 54 – 87 – 131?	Yes	186		15.4	26.0	33.4	4.07	0.42

(*Continued*)

Table 2. Continued.

Category	#	Task	Prompt	AI correct?	Response (shortened)		CTR response time (sec., n = 113)			Difficulty *(1–10, n = 45)*	
					Correct	ChatGPT	p_{30}	Median	Mean	Mean	Std. error
Brainteaser	25	bat_and_ball	A bat and a ball cost $$1.10 in total. The bat costs $$1.00 more than the ball. How much does the ball cost?	Yes	ball 0.05$$, bat 1.05$$		17.9	30.4	41.1	2.38	0.27
	26	5_machines	If it takes 5 machines 5 min to make 5 widgets, how long would it take 100 machines to make 100 widgets?	**No**	5 min	20 min	21.8	32.9	49.4	2.78	0.34
	27	roses_on_lake	In a lake, there is a patch of lily pads. Every day, the patch doubles in size. If it takes 48 days for the patch to cover the entire lake, how long would it take for the patch to cover half of the lake?	Yes	47 days		13.0	21.6	30.9	3.16	0.34
	28	gallon_bucket	You have a three-gallon bucket and a five-gallon bucket. How do you measure out exactly four gallons?	Yes	Description of procedure to solve the task		24.8	41.2	58.8	4.87	0.39
Text-editing	29	anagrams	What are all possible anagrams of the word 'dome', including those that are not words with any meaning?	**No**	4! = 24 possible anagrams	Listed only 12 anagrams	13.4	19.2	30.0	4.80	0.38
	30	replace_words	In the following sentence, replace each first name with a separate number: 'Peter, Paul, Mary, John, and Mandy are going to the park to visit Neil by the lake. Paul and Neil want to play basketball while Mandy and Peter are indifferent, and Mary and John just want to hang out with the others.'	Yes	Adjusted sentence		28.1	45.0	51.0	3.18	0.38
	31	haiku	Create a haiku on AI.	Yes	Structurally / thematically correct haiku		9.1	12.0	20.8	5.33	0.47
	32	order_list	Please order this list alphabetically in ascending order: Robot, Radiant, River, Renaissance, Reindeer, Romantic, Rocket, Rustic, Ruby, Razor, Rainbow, Rhythm, Rusty, Ravioli, Rider, Rollercoaster, Rhino, Roulette, Rug, Ribbon	Yes	Sorted list		17.9	28.8	34.7	2.62	0.27
Code	33	200px_rainbow	Please provide a snippet of code that will create a 200-pixel wide rainbow in HTML.	Yes	Working code snippet		11.2	15.5	21.2	7.13	0.46
	34	recursive_function	Create a recursive function to sort a list of words alphabetically.	Yes	Working function		9.6	14.7	21.6	7.02	0.43
	35	palindrom_function	Write a js function to detect a palindrome in a sentence.	Yes	Working function		8.9	13.1	18.9	7.64	0.41
	36	code_analysis	The following is Python code: i = 1 ar = [1,2,3] i,ar[i] = 0,5 What is now the value of ar?	**No**	ar = [5, 2, 3]	ar = [1, 5, 3]	9.6	14.4	27.8	6.80	0.48

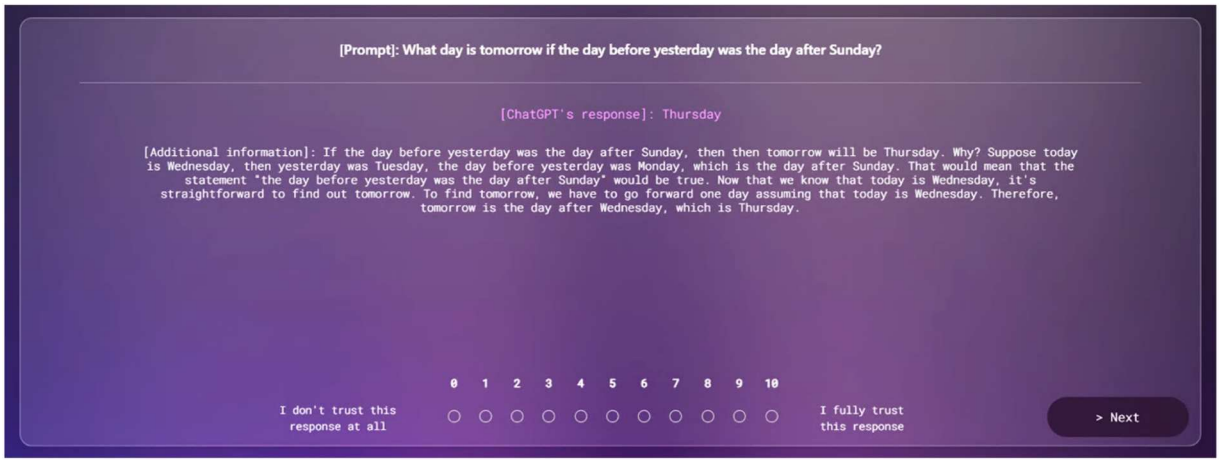


Figure 2. Screenshot of task presentation including response scale.

the *optimal* level (i.e. **0** for incorrect and **10** for correct responses).[3] To illustrate this: A participant who indicated a willingness to rely of 4 for an incorrect response would receive a performance score of 6 (i.e. 10 – |4–**0**|). Similarly, a participant who indicated a willingness to rely of 9 for a correct response would receive a performance score of 9 (i.e. 10 – |9–**10**|). The bonus payment was directly inferred from the overall *performance score* across all 36 tasks where the highest possible total score (36 × 10 = 360) yielded the maximum payoff. The distribution of willingness to rely and performance per category as well as per task is shown in Appendix B.

3.5. Survey

After completing the actual experiment, we surveyed several control variables. Specifically, we surveyed participants' *general risk propensity* (Dohmen et al. 2011), *disposition to trust in technology* (Merritt et al. 2013), *technological literacy,* as well as *experience* with, and *usage* of ChatGPT specifically. The full survey is available in Appendix C.

3.6. Manipulation check

To confirm that our treatment manipulation was successful, we used a single-item manipulation check ('*I felt a lot of time pressure to evaluate the AI's responses*', 7-point Likert scale from 1 to 7). Participants from the time pressure treatment scored significantly higher on this scale ($MC_{TP} = 6.16 > 3.40 = MC_{CTR}$; $p < .001$).

3.7. Realism check

To confirm that our experiment was (1) perceived plausible overall and (2) similarly so across both treatment conditions, we used a two-item realism check ('*I think that the provided prompts and responses were a good representation of text-generative AI performance*' and '*The scenario presented here (i.e. reading and assessing AI text responses) seems plausible in general and may be a common part of future knowledge-work*', 7-point Likert scales from 1 to 7). Overall, participants assessed the AI performance as very realistic ($RC_1 = 5.31$) and representative of future knowledge work ($RC_2 = 5.36$) with no significant differences between treatment conditions ($p_1 = .056$; $p_2 = .538$). The high average values >5 for both checks indicate participants predominantly selected the two highest possible levels of agreement with the measures. The realism and manipulation checks are plotted in Appendix D.

3.8. Procedure and sample

We recruited 240 participants through the online research platform Prolific. Within the field of IS research, Prolific is accepted as a reliable and high-quality source of participants with markedly higher data quality as compared to other platforms (e.g. MTurk). This is evidenced by participants passing attention checks more consistently and providing more meaningful answers (Douglas, Ewell, and Brauer 2023; Peer et al. 2017, 2021). Using this platform also mitigates self-selection bias and enhances generalizability by tapping into a diverse pool of participants (Russo 2022).

We used a gender-balanced sample of participants with English language fluency. The treatment group was recruited after the control group so that we were able to adjust the time limits according to the completion times observed in the control group. To mitigate potential confounding effects, invitations in both conditions were sent at the same time of day. Note that participants from the control group could not participate in the treatment group again (i.e. between-subjects design).

3.9. Attention check

To ensure high data quality, we employed four separate attention checks across the different stages of the experiment (briefing, main experiment, and post-experiment survey). The final dataset contains 228 participants, after 12 submissions with more than one failed attention check were excluded ($N_{CTR} = 113$; $N_{TP} = 115$).

Participants' age ranged from 18 to 71 years (mean = 29.2, median = 26; standard deviation = 9.08). The sample exhibits a balanced gender split (male = 112, female = 116). In terms of geographic distribution, 58% of participants were from Europe, 34% from Africa, and 8% from the Americas, Asia, and the rest of the world. Most of the participants (51%) reported being employed – either in full-time employment (38%), part-time (11%), or due to start a new job shortly (2%) – seeking a job (21%), not being in paid work (12%), or did not provide employment information (16%).

Overall, the median completion time was 20.7 min and participants received a median payoff of GBP 3.22 (including both fixed and performance-contingent components, yielding a compensation of GBP 9.33 / hour). Note that, by design of using time constraints in one condition, expected completion times differ across treatments. To account for this, we adjusted the fixed payment accordingly, ensuring adequate compensation for participants (Table 3).

3.10. Randomisation check

To check whether the randomisation worked (i.e. randomly allocating participants into the control or treatment groups), we conduct a randomisation check. For this, we use a set of linear and logit regression models with age, gender, trust, and skepticism towards AI before the experiment, ChatGPT usage time, and the manipulation check for time pressure as dependent variables (Table 4). The treatment variable (time pressure: 0 = no, 1 = yes) does not have any significant effects in any of these models except the manipulation check for time pressure, where it is expectedly highly significant. We thus conclude that our treatment randomisation was successful.

3.11. Consent, ethical approval, and data management

Prior to participation, all participants were presented with an informed consent page, which emphasised the voluntary nature of the study, the anonymous data handling, and their right to withdraw from the experiment at any time. This study was conducted in full compliance with our institution's ethical guidelines, with all involved researchers having completed training and certification in ethical research design and human subject research. Before commencing the

Table 3. Compensation for participants.

	Control group (n = 113)	Treatment group (n = 115)
Fixed Payment	GBP 3.00	GBP 2.25
Bonus Payment (med.)	GBP 0.63	GBP 0.57
Time (tasks only, med.)	15 min 54 s	9 min 18 s
Time (full experiment incl. survey, med.)	26 min 4 s	17 min 41 s
Overall payment per hour (med. pay / med. time)	GBP 8.36 / h	GBP 9.54 GBP / h
	GBP 9.33 / h	

Table 4. Randomisation check.

	Dependent variable					
	Age	**is Female**	**Trust in AI**	**AI Skepticism**	**ChatGPT Usage**	**MC Time Pressure**
	OLS	*logit*	*OLS*	*OLS*	*OLS*	*OLS*
	(1)	(2)	(3)	(4)	(5)	(6)
Time Pressure (0 = no, 1 = yes)	1.883	0.036	0.164	0.157	−0.015	2.758***
	(1.198)	(.265)	(.176)	(.215)	(.221)	(.193)
Constant	28.248***	−0.053***	3.071***	2.487***	1.850***	2.398***
	(.851)	(.188)	(.125)	(.153)	(.157)	(.137)
Observations	228	228	228	228	228	228
R^2	0.011		0.004	0.002	<0.001	0.476
Adjusted R^2	0.006		−0.001	−0.002	−0.004	0.473
Log Likelihood		−157.993				
Akaike Inf. Crit.		319.987				
Residual Std. Error (df = 226)	9.048		1.331	1.625	1.667	1.455
F Statistic (df = 1; 226)	2.468		0.866	0.531	0.004	204.910***

Note: Standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

study, we obtained ethical approval from our institution's *Ethical Research Board* (approval #20230530). The review also covered the data management plan for this project.

4. Results

4.1. Human-AI-team performance

First, we investigated the effect of time pressure, AI advice faultiness, and task complexity on human-AI-team *performance*. Figure 3 shows human-AI-team performance by time pressure (yes/no) and (1) AI response faultiness (correct/incorrect) and (2) task complexity (0–10):

The charts indicate marked and consistent decreases in performance under time pressure. Specifically, participants seem to struggle with identifying faulty AI responses as such, whereas identifying correct responses causes less trouble. Time pressure seems to exacerbate this deterioration, with lower performance observed for both correct and faulty AI responses but with an additional loss for faulty ones. As expected, task complexity appears to have a slightly negative effect on performance both with and without time pressure.

To corroborate our visual assessment statistically, we conducted a set of linear mixed effect regressions, addressing the data structure with repeated measures per participant (i.e. n = 228 participants × 36 tasks = 8,208). Specifically, we used random intercepts on the participant level. To understand the relationship between the dependent variable *performance* and task complexity, AI response faultiness, and the presence of time pressure, we gradually introduced the main treatment and control variables to the models and control for age, sex, and period (Table 5).

In the first model (I), we only used the treatment variable (time pressure) and see a significant negative effect on human-AI-team performance ($\beta = -0.693$, $p < .001$), supporting $\mathbf{H_1}$ (*The presence of time pressure adversely affects human-AI-team performance*). When task complexity and AI response faultiness are included in the model (II), the effect of time pressure remains, and the negative effects of task complexity ($\beta = -0.309$, $p < .001$) and AI response faultiness ($\beta = -3.654$, $p < .001$) emerge. These findings reaffirm $\mathbf{P_1}$ (*Increased Task Complexity adversely affects human-AI-team performance)* and $\mathbf{P_2}$ (*Increased AI response faultiness adversely affects human-AI-team performance),* respectively. These effects remain stable and significant once the control variables age, gender, and period are introduced to the model (III), and these variables do not exert any significant effects.

The inclusion of interaction effects in the model (IV) uncovered a more nuanced view. Time pressure exhibits a significant interaction effect with both task complexity ($\beta = 0.374$, $p < .001$) and AI response faultiness ($\beta = -0.561$, $p < .001$). This negative interaction effect between time pressure and faulty AI responses indicates that time pressure exacerbates the negative impact of AI errors on human-AI-team performance, supporting $\mathbf{H_3}$ (*Time pressure amplifies the adverse effect of AI response faultiness on human-AI-team performance).*

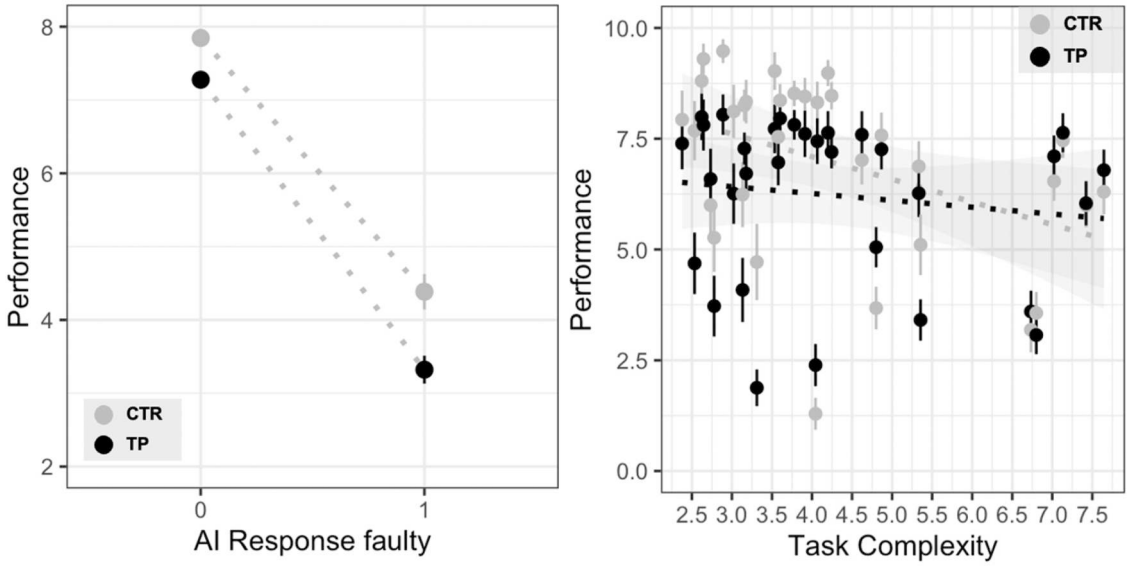


Figure 3. Human-AI-team performance by time pressure and (1) AI response faultiness (left) and (2) task complexity (right) for control (CTR) and time pressure (TP) groups. Error bars show 95% confidence intervals.

Table 5. Regression of human-AI-team performance, time pressure, task complexity, and AI response faultiness

	DV: Human-AI-team performance				
	I	II	III	IV	V
Time Pressure	**−.693*****	**−.693*****	**−.690*****	**−2.145*****	**−2.141*****
(0 = no, 1 = yes)	(.099)	(.099)	(.100)	(.212)	(.213)
Task Complexity		**−.309*****	**−.308*****	**−.498*****	**−.497*****
		(.022)	(.022)	(.031)	(.031)
AI response faultiness		**−3.654*****	**−3.661*****	**−3.371*****	**−3.380*****
(0 = no, 1 = yes)		(.076)	(.076)	(.108)	(.108)
Time Pressure ×				**.374*****	**.374*****
Task Complexity				(.044)	(.044)
Time Pressure ×				**−.561*****	**−.556*****
AI response faultiness				(.152)	(.152)
Control					
Period			.005		.005
(from 1 to 36)			(.003)		(.003)
Sex			.109		.109
(female = 0, male = 1)			(.099)		(.099)
Age			−.001		−.001
			(.006)		(.006)
Constant	**6.981*****	**9.208*****	**9.083*****	**9.940*****	**9.822*****
	(.070)	(.118)	(.210)	(.151)	(.230)
# Observations	8,208	8,208	8,208	8,208	8,208
Marginal R^2	.01	.24	.23	.247	.248
Conditional R^2	.029	.266	.276	.273	.274
Log Likelihood	−21,835.400	−20,724.500	−20,733.060	−20,686.030	−20,694.770
Akaike Inf. Crit.	43,678.800	41,460.990	41,484.120	41,388.070	41,411.540
Bayesian Inf. Crit.	43,706.850	41,503.070	41,547.240	41,444.170	41,488.680

Note: Standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

Interestingly, the interaction between time pressure and task complexity is positive, indicating that task complexity plays a smaller role in human-AI-team performance under time pressure. While $\mathbf{H_2}$ anticipated a compounding negative effect, the positive coefficient ($\beta = 0.374$, $p < .001$) implies that under time pressure, the adverse impact of task complexity is *less* pronounced. These effects also remain stable and significant once the control variables age, gender, and period are introduced to the model (V). Figure 4 summarises our findings on human-AI-team performance.

The inclusion of various individual difference measures from the post-experiment survey (see Appendix C) – including risk propensity, technology proficiency, familiarity and usage of ChatGPT, trust in technology, and attitudes towards text-generative AI before and after the experiment – revealed neither significant direct effects nor interaction effects with time pressure on performance.

4.2. Decision strategies and choice univocity

Next, we explored how performance and decision behaviour between treatment groups vary and why this might be the case. As we have argued in Section 2, time pressure might cause participants to shift away from a (more) systematic approach and follow heuristic strategies. Specifically, they might gauge the AI's overall performance based on individual preconceptions, from what they picked up in the media, or first-hand experience with tools such as ChatGPT. Rather than judging

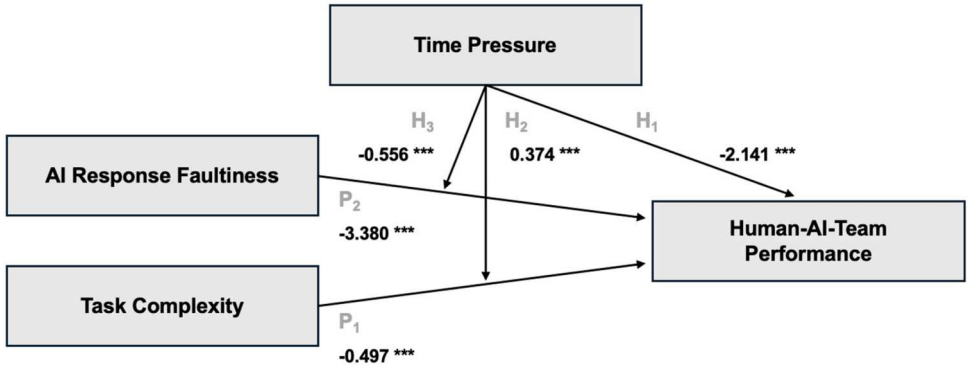


Figure 4. Research Model including coefficients for observed effects.

each task individually and making a distinct decision, they might provide more vague responses, attempting to reflect the AI's overall performance. For instance, if a participant believes the AI's performance to be 75%, they might not (deliberately) attempt to identify the 27 correct and the 9 incorrect responses (36 in total) – and clearly label them as such – but rate all responses with a score of (around) 75% to reflect their general assessment of the AI's performance.

While both 'strategies' would – on average – yield similar levels of willingness to rely (i.e. 75%), they would differ concerning *performance* and the frequency of *choice univocity* (i.e. a stated willingness to rely of 0 or 10).[4] For instance, a deliberate strategy involves explicitly labelling each response as correct or incorrect, while a heuristic approach results in vaguer judgments.

Importantly, both described 'strategies' represent idealised scenarios and actual behaviour will likely be more incoherent (e.g. as participants might employ a mix of both heuristic regimes, adapt their approach over the course of the experiment, or build on specific own knowledge). Regarding performance, the perfect *deliberate strategy* would identify all correct and incorrect responses and label them as such, consequently yielding a performance of 100%. The *heuristic strategy* would (given that 75% of all AI responses are correct), yield a performance of $75\% \times 75\% + 25\% \times 25\% = 62.5\%$. Likewise, the perfect deliberate strategy would yield a choice univocity rate of 100%, that is, labelling all responses as either correct or incorrect – while the heuristic approach would yield zero choice univocity.

Figure 5 illustrates where the average performance and choice univocity of both groups fall along the spectrum of these two idealised strategies. The treatment group (i.e. with time pressure) achieves an average performance of 62.5%, precisely as assumed in our heuristic strategy model. In contrast, the control group, unencumbered by time constraints, achieves higher performance of around 70%. Notably, despite the lack of explicit information about the AI's reliability, the treatment group's results align with the anticipated outcomes of heuristic decision-making. Furthermore, the control group demonstrates less frequent hedging of their responses and provides more definitive judgments about their willingness to rely on the AI. Overall, the observed performance and decision patterns across both groups align with theoretical expectations.

Figure 6 merges the metrics of average choice univocity and average performance for both strategies in one visualisation. It is important to note that the potential performance range for 100% choice univocity spans from 0 (consistently misjudge all tasks) to 10 (consistently judge all tasks correctly). In contrast, the performance range for 0% choice univocity is bounded between a minimum of 1 (consistently misjudging all tasks but avoiding choice univocity by 1 point) and a maximum of 9 (consistently judging all tasks correctly, but avoiding choice univocity by 1 point). While average performance is concentrated in the range from 5 to 8 points, choice univocity is spread almost evenly across the range from 0 to 100%.

A deeper look at choice univocity uncovers that while participants in the treatment group (i.e. under time pressure) on average make notably fewer definite choices over the experiment's entirety, this difference decreases considerably over time (Figure 7).

Separating choice univocity further into *fully trust (10)* and *do not trust at all (0)* reveals the source of the underlying effect (Figure 8). In the control group, time already spent in the experiment had no observable effect on participants' willingness to rely on the AI. Both possible frequencies of choice univocity remain stable throughout, with *fully trust* (10) selected for 41.5% of tasks and *do not trust at all* (0) selected for 9.4% of tasks.

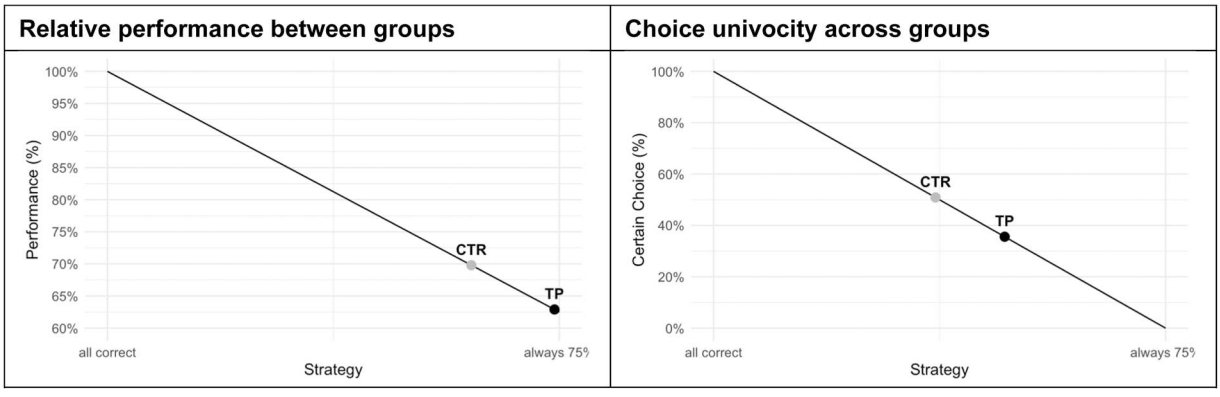


Figure 5. Avg. performance and share of choice univocity (i.e. willingness to rely = 0 or 10) for groups CTR and TP, on the spectrum between strategies always choose the correct response and always respond with general trust in AI (i.e.75%).

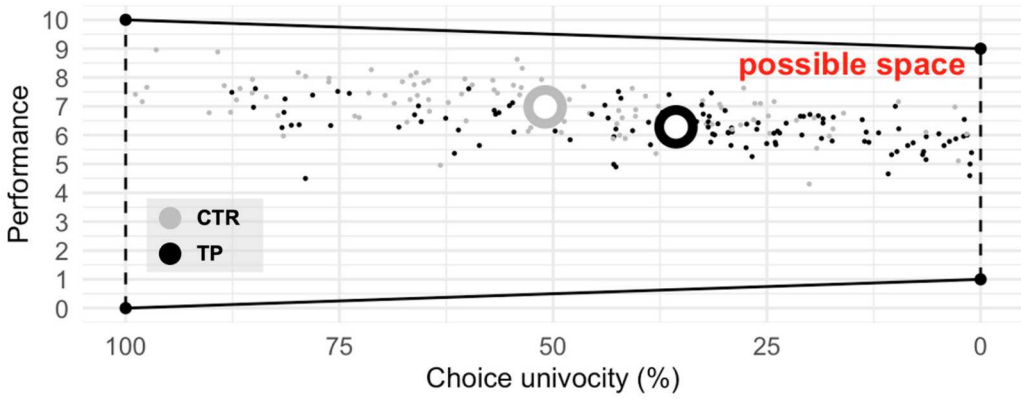


Figure 6. Avg. performance and choice univocity (i.e. willingness to rely = 0 or 10) for groups CTR and TP, group averages as bold circles.

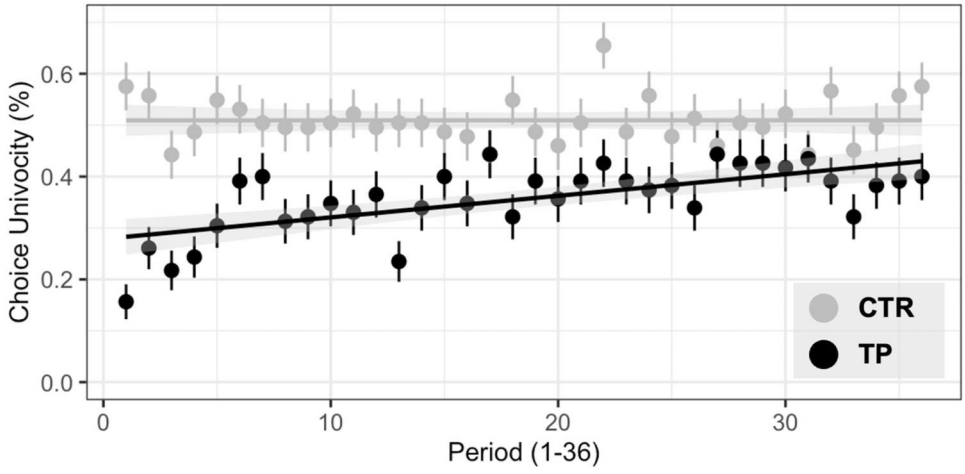


Figure 7. Choice univocity by period of a task for groups CTR and TP.

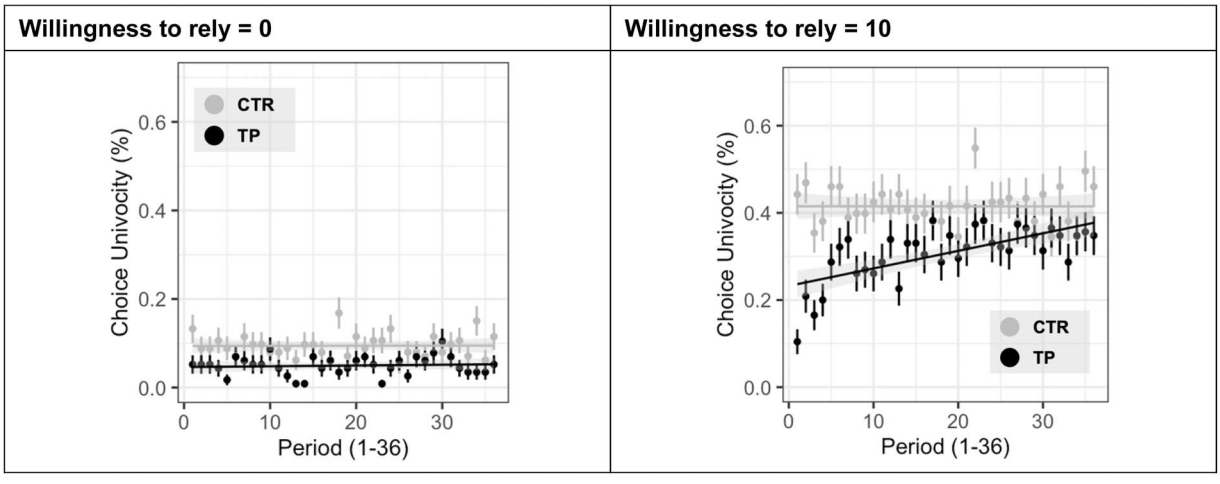


Figure 8. Distribution of no / full willingness to rely by period of the task for groups CTR and TP.

For participants in the treatment group, we also observed a stable, yet lower rate of *choice univocity* for *do not trust at all* (0). The entire increase in choice univocity can be attributed to a steady and substantial increase in stated willingness to *fully trust* (10) as the experiment progressed.

4.3. Tendency to commit to definite choices

To quantify these visual assessments, we conducted a set of mixed effect regressions with the dependent variable *choice univocity* (Table 6). Again, we gradually introduced the main treatment and control variables to the models and control for age, sex, and period.

Time pressure exerts a large and significant negative impact on choice univocity across models, with coefficients ranging from $\beta = -.153$ for models not accounting for interaction effects to $\beta = -.391$ for models accounting for interaction effects with task complexity (all p-values < .001). Similarly, task complexity has a significant effect across models (all p-values < .001), with smaller coefficients ($\beta = -.084$) for models II and III not accounting for interaction effects with time pressure, and a larger coefficient ($\beta = -.112$) for models IV and V accounting for such effects. AI response faultiness – a binary variable – has a comparably small effect size that is only significant at the 5%-threshold in model III. Under time pressure, a significant positive interaction effect exists with task complexity ($\beta = .056$, $p < .001$), indicating that under time pressure, participants are more likely to make a univocal choice regarding their reliance on the AI. This positive interaction effect offsets exactly half of the negative effect of task complexity on its own. There is no significant interaction between time pressure and AI response faultiness.

Neither age nor sex exhibit significant effects on choice univocity, however, period has a significant and positive effect ($\beta = .002$, $p < .001$). As the experiment progresses, participants appear to increase their frequency of making definitive statements regarding their willingness to (not) rely on the AI. All models are shown in Table 6.

4.4. Participants' willingness to rely on AI responses

Finally, we visually investigated the relationship between participants' *willingness to rely on the AI*, AI response correctness, and task complexity, once again differentiated by the time pressure treatment (Figure 9).

Here, the effects are less straightforward. In the absence of time pressure, participants exhibit a markedly higher willingness to rely on correct responses than on faulty ones, indicating participants' general ability to process cues and differentiate between

Table 6. Regression of choice univocity, time pressure, task complexity, and AI response faultiness.

	DV: Choice univocity				
	I	II	III	IV	V
Time Pressure	**−.153*****	**−.153*****	**−.154*****	**−.390*****	**−.391*****
(0 = no, 1 = yes)	(.031)	(.031)	(.031)	(.041)	(.041)
Task Complexity		**−.084*****	**−.084*****	**−.112*****	**−.112*****
		(.003)	(.003)	(.004)	(.004)
AI response faultiness		−.020	**−.022***	−.018	−.021
(0 = no, 1 = yes)		(.011)	(.011)	(.015)	(.015)
Time Pressure ×				**.056*****	**.056*****
Task Complexity				(.006)	(.006)
Time Pressure ×				−.004	−.002
AI response faultiness				(.021)	(.021)
Control					
Period			**.002*****		**.002*****
(from 1 to 36)			(.0004)		(.0004)
Sex			.053		.053
(female = 0, male = 1)			(.031)		(.031)
Age			.001		.001
			(.002)		(.002)
Constant	**.509*****	**.871*****	**.787*****	**.991*****	**.906*****
	(.022)	(.026)	(.058)	(.029)	(.059)
# Observations	8,208	8,208	8,208	8,208	8,208
Marginal R^2	.024	.089	.094	.096	.101
Conditional R^2	.228	.295	.298	.303	.306
Log Likelihood	−5,098.310	−4,743.989	−4,747.817	−4,709.291	−4,713.213
Akaike Inf. Crit.	10,204.620	9,499.978	9,513.635	9,434.583	9,448.427
Bayesian Inf. Crit.	10,232.670	9,542.056	9,576.750	9,490.685	9,525.568

Note: Standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

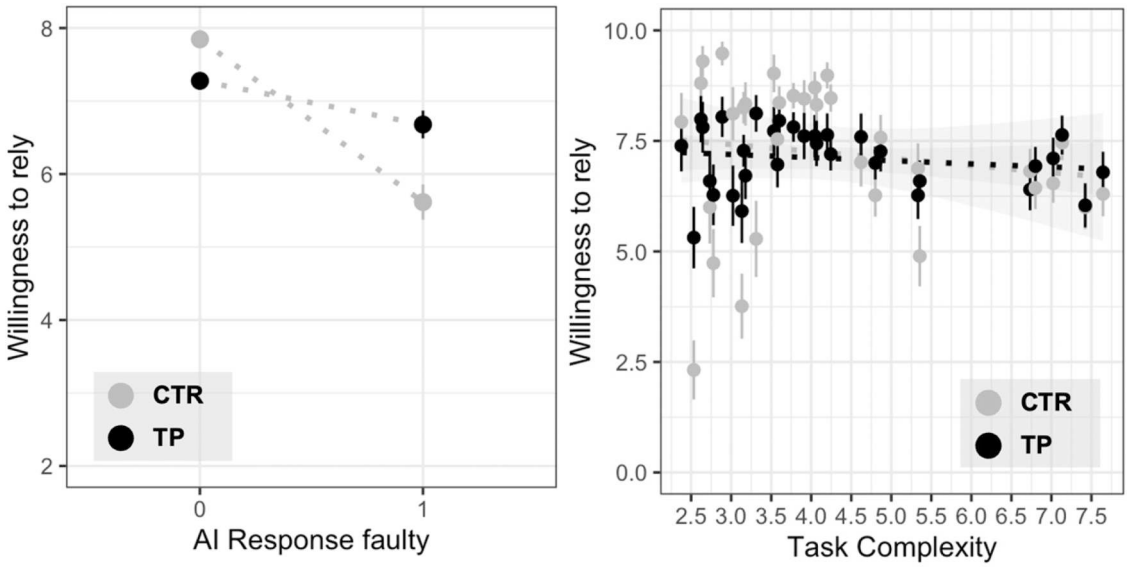


Figure 9. Willingness to rely on AI response by time pressure and (1) AI response faultiness (left) and task complexity (right) for groups CTR and TP. Error bars show 95% confidence intervals.

accurate and faulty responses. When time pressure is introduced, however, there is no uni-directional response such as trusting the AI more (or less) overall. Remarkably, participants will rely less on correct but more on incorrect responses with only a small differential in between. This convergence of trust levels is a signal for deterioration of the ability to differentiate between correct and faulty AI responses, potentially due to a switch in the underlying cognitive processing from deliberate towards heuristic reasoning. Recall that participants did not face a binary decision to either fully or not-at-all rely on the AI response, but could indicate their willingness to rely on the AI on a continuum from 0 to 10. Task complexity appears to have

Table 7. Regression of participants' willingness to rely on AI, time pressure, task complexity, and AI response faultiness.

	DV: Willingness to rely on AI response				
	I	II	III	IV	V
Time Pressure	−.162	−.162	−.170	**−.914*** **	**−.921*** **
(0 = no, 1 = yes)	(.141)	(.141)	(.142)	(.232)	(.232)
Task Complexity		**−.108*** **	**−.105*** **	**−.150*** **	**−.146*** **
		(.022)	(.022)	(.031)	(.030)
AI response faultiness		**−1.388*** **	**−1.412*** **	**−2.204*** **	**−2.240*** **
(0 = no, 1 = yes)		(.075)	(.075)	(.106)	(.106)
Time Pressure ×				.082	.080
Task Complexity				(.043)	(.043)
Time Pressure ×				**1.618*** **	**1.638*** **
AI response faultiness				(.149)	(.149)
Control					
Period			**.020*** **		**.021*** **
(from 1 to 36)			(.003)		(.003)
Sex			−.147		−.147
(female = 0, male = 1)			(.141)		(.141)
Age			.004		.004
			(.008)		(.008)
Constant	**7.289*** **	**8.096*** **	**7.689*** **	**8.476*** **	**8.055*** **
	(.100)	(.137)	(.275)	(.165)	(.290)
# Observations	8,208	8,208	8,208	8,208	8,208
Marginal R^2	.001	.041	.046	.054	.059
Conditional R^2	.089	.13	.135	.143	.149
Log Likelihood	−20,860.96	−20,681.07	−20,669.79	−20,622.72	−20,609.71
Akaike Inf. Crit.	41,729.92	41,374.14	41,357.58	41,261.44	41,241.43
Bayesian Inf. Crit.	41,757.97	41,416.22	41,420.70	41,317.54	41,318.57

Note: Standard errors in parentheses. *$p < .05$; **$p < .01$; ***$p < .001$.

only a small negative effect on participants' willingness to rely on AI responses. We applied an analogous set of prior regressions for *choice univocity* and *performance* to explain participants' stated *willingness to rely* on the displayed AI responses (Table 7).

Interestingly, time pressure does not have any significant *overall* effect on participants' willingness to rely on the AI's responses (models I, II, and III). However, we see negative effects of both task complexity and AI faultiness. Specifically, incorrect AI responses consistently and significantly reduce willingness to rely across all models (all p-values < .001). Similarly, it also decreases with increasing task complexity (all p-values < .001), When introducing the control variables, we see that – again – *period* has a significant and positive effect. We interpret this as due to habituation, where participants get more comfortable in relying on the AI throughout the experiment (i.e. from periods 1 to 36).

Models IV and V provide strong statistical support for the interaction of time pressure and AI response faultiness as shown in Figure 9 above. Under time pressure, participants tend to lend higher (lower) credence to incorrect (correct) responses as compared to the control group without time pressure. Curiously, there is no significant interaction effect of task complexity and time pressure, indicating that participants might not consciously process the task and employ a more heuristic approach towards their willingness to rely.

4.5. Mediation analysis

Lastly, we conducted a mediation analysis to evaluate whether the effect of time pressure on performance is mediated by willingness to rely. We fitted two linear mixed-effects models to estimate both the direct and indirect pathways (Figure 10, Table 8). The first model tested the effect of time pressure (independent variable) on willingness-to-rely (mediator), incorporating a random intercept for participants to account for within-subject variability across the experiment. The second model predicted performance (dependent variable) as a function of time pressure and willingness-to-rely, again including random intercepts for participants. We then applied a standard mediation framework to compute the (indirect) *Average Causal Mediation Effect* (*ACME*), the *Average Direct Effect* (*ADE*), and the *Total Effect*. The (indirect) mediation effect represents the pathway from time pressure to performance through willingness-to-rely, while the direct effect quantifies the remaining relationship between time pressure and performance after accounting for willingness-to-rely. The total effect is the sum of these two pathways.

As expected from our linear regression model on performance, the results confirm a significant total effect of time pressure on performance, with an estimated coefficient of $\beta = -0.696$ (CI95: [−0.862, −0.520], $p < .001$). The direct effect, which bypasses the mediation pathway, was similarly significant (estimate: $\beta = -0.628$;

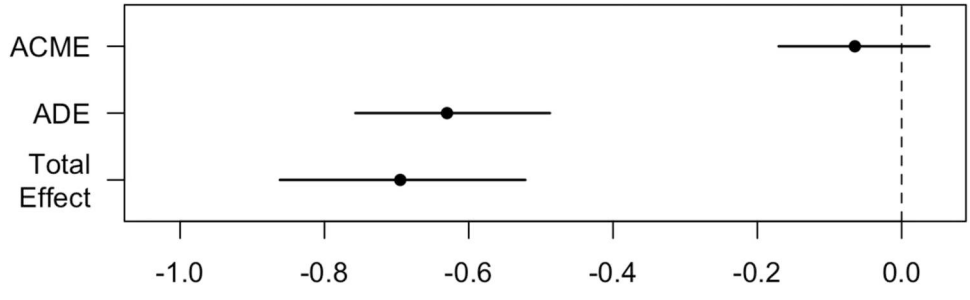


Figure 10. Results of mediation analysis.

Table 8. Causal Mediation Analysis – Quasi-Bayesian confidence intervals.

	Mediator groups:	Prolific ID	**Sample size:**	8,208
	Outcome groups:	Prolific ID	**Simulations:**	500
		Output based on overall averages across groups		
Effect	Estimate	95% CI Lower	95% CI Upper	*p*-value
ACME (Average Causal Mediation Effect)	−.0648	−.1704	.04	.22
ADE (Average Direct Effect)	**−.6301**	**−.7570**	**−.49**	**<.001*** **
Total Effect	**−.6949**	**−.8617**	**−.52**	**<.001*** **
Proportion mediated	.0912	−.0662	.22	.22

Note: *$p < .05$; **$p < .01$; ***$p < .001$.

CI95: [−0.776, −0.490], $p < .001$). This suggests that time pressure exerts a strong independent influence on performance, potentially by triggering a switch towards more heuristic reasoning.

In contrast, the mediation effect through willingness-to-rely was small and not statistically significant ($\beta = -0.067$; CI95: [−0.175, 0.050], $p = .27$). The confidence interval includes zero, indicating no conclusive evidence that willingness-to-rely mediates the relationship between time pressure and performance. This finding suggests that, while willingness-to-rely may be relevant to performance in certain contexts, it does not play a meaningful role in linking time pressure to performance in this experiment. The dominance of the direct effect in the total effect thus underscores the direct pathway as the primary mechanism by which time pressure influences performance.

5. Discussion

Our literature review indicated that human-AI collaborative performance is negatively influenced by time pressure, AI system reliability, and task complexity. These effects were anticipated to occur independently and to intensify when combined. However, our findings challenge this assumption and reveal an unexpected interaction between task complexity and time pressure, which we discuss in the following.

5.1. Summary of results

As expected, our study confirms the stated premises (negative effect of task complexity ($\mathbf{P_1}$) and AI advice faultiness ($\mathbf{P_2}$) on human-AI-team performance; both p-values < .001). Faced with more challenging tasks or faulty AI responses, participants were less able to gauge the AI response's accuracy. Furthermore, our experiment supported our hypothesis on the negative effect of time pressure on human-AI-team performance ($\mathbf{H_1}$; $p < .001$). Moreover, we found this effect to intensify further for tasks concerning faulty AI responses, supporting our third hypothesis on the interaction effect ($\mathbf{H_3}$). Interestingly, our hypothesis that time pressure amplifies the adverse effect of task complexity on human-AI-team performance ($\mathbf{H_2}$) was not supported by the data; instead, the presence of time pressure in fact offsets the negative effect of task complexity by about 75%. These results suggest that, while time pressure seems to generally undermine decision quality and performance, its influence may partially offset the detrimental effects of task complexity. Our mediation analysis confirmed the direct effect of time pressure on performance independent of participants' willingness to rely, indicating a negligible mediation effect.

5.2. Theoretical contributions

Our findings contribute to the understanding of dual-processing theories in the context of human-AI collaboration. Traditionally, such models posit a direct and binary shift from analytical (System 2) to heuristic (System 1) processing under stress. However, our results suggest a more nuanced effect: time pressure does not uniformly degrade cognitive evaluation but modulates the trade-off between speed and accuracy in a task-dependent manner. This finding necessitates a reconsideration of how time constraints are integrated into models of human decision-making, particularly in AI-supported environments where cognitive processes must deal with algorithmically generated suggestions, advice, text fragments, and so forth.

While our theoretical framing draws primarily on dual-processing theory and the HSM, it is important to recognise that such models have been critiqued for their binary assumptions and limited scope in explaining decision-making under real-world constraints (Evans, St, and Stanovich 2013). Our results suggest a need to integrate perspectives from models of cognitive flexibility and learned heuristics (e.g. Marewski and Gigerenzer 2012), which posit that individuals adaptively select decision strategies based on environmental cues, task demands, and prior experience.

In particular, the observed offsetting effect of time pressure on task complexity supports interpretations aligned with bounded rationality (Simon 1955) and resource-rational models of cognition (Lieder and Griffiths 2020), which argue that decision strategies are shaped not only by cognitive limitations but by an optimisation of cognitive resource use in context-sensitive ways. These perspectives move beyond the rigid System 1 vs. 2 dichotomy and suggest that heuristics can be both learned and flexibly deployed – potentially explaining why participants under time pressure resorted to simplified, but not necessarily indiscriminate, processing strategies.

To offer a more complete theoretical account, we refer to models of adaptive cognition such as fast-and-frugal heuristics (Gigerenzer and Gaissmaier 2011) and satisficing behaviour (Simon 1955). These models suggest that under time pressure, individuals shift to simplified decision rules that limit cognitive load by focusing on the most salient diagnostic information to arrive at a 'good enough' basis of information to come to a decision. As a result, performance is mostly

negatively affected by time-pressure and becomes less sensitive to the complexity of the task, attenuating the expected negative effects of high task difficulty.

Moreover, incorporating literature on emotional and stress-induced factors (Lerner and Keltner 2001; Starcke and Brand 2012) could further elucidate how emotional states such as anxiety or stress under time pressure, might moderate dual-processing. Additionally, metacognitive awareness (Ackerman and Thompson 2017; Flavell 1979) might explain whether participants consciously perceived their shift from analytical to heuristic reasoning.

Overall, time pressure had a significant negative effect on performance (−2.141; $p < .001$). However, when considering the interaction between task complexity and time pressure (−0.123, $p < .001$), the relative impact of task complexity was approximately 75% smaller than in the absence of time constraints (−0.497, $p < .001$). This outcome is counterintuitive, challenges prevailing cognitive load theories (Kahneman and Klein 2009) and warrants further investigation into the underlying mechanisms as conventional reasoning suggests that heuristic decision-making induced by time pressure and task complexity should exert compounding negative effects, widening the performance gap between the control and treatment groups. Instead, our findings reveal a narrowing of this gap as task complexity increases, as illustrated in Figure 11. Such findings are consistent with the notion that satisficing and heuristic strategies employed under time pressure effectively reduce individuals' sensitivity to task complexity by bounding the search process and limiting deliberation (Payne, Bettman, and Johnson 1993). Nevertheless, it is important to note that the control group consistently outperformed the time pressure group across all levels of task complexity, and time pressure did not result in sustained superior performance at any stage of the experiment.

5.3. Ex-post reasoning

Time pressure appears to induce a fundamental shift in human decision-making when dealing with AI-generated responses. Under time pressure, task complexity barely affects performance. Conversely, in the absence of time pressure, we observe the expected negative effect. While we did not observe significant effects of time pressure on AI adherence, one potential explanation is that, under time pressure, individuals cognitively disengage from the task itself, shifting attention primarily to the AI response's perceived trustworthiness. Rather than consciously adjusting their trust levels, participants maintained stable trust, suggesting the underlying cognitive mechanism involves reduced analytical engagement and increased reliance on a baseline level of trust in the AI overall. Figure 12 below illustrates the significant decline in univocal choices as task complexity increases and the decreased overall choice univocity under time pressure.

Both *sensitivity* (i.e. trust-based performance or the proportion of correct AI responses accurately identified as correct by humans) and *specificity* (i.e. distrust-based performance or the proportion of incorrect AI responses accurately identified as incorrect by humans) decline under time pressure. However, this deterioration is particularly pronounced for sensitivity (see Appendix F). In the absence of explicit performance cues, participants exhibited a high baseline level of willingness to rely upon the AI system. While time pressure initially reduced both participants' willingness to rely and their tendency for choice univocity, skepticism was only temporary. Throughout the experiment,

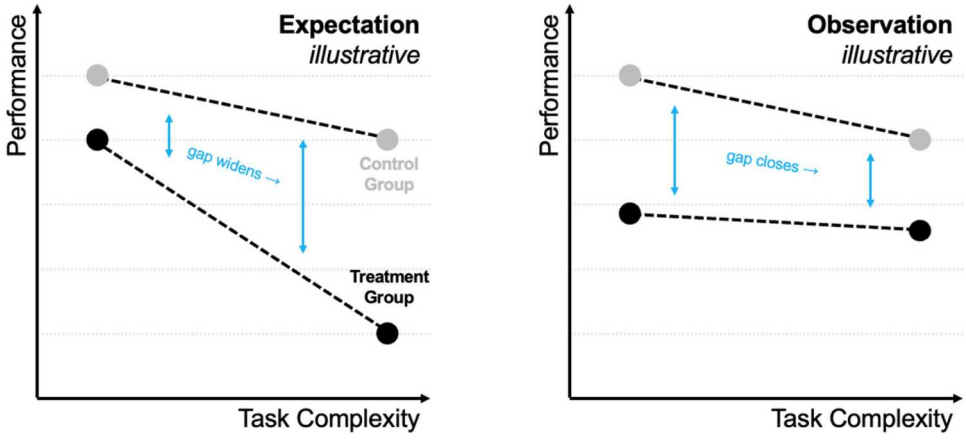


Figure 11. Illustration of expected vs. observed effect directions for control group (no time pressure) and treatment group (time pressure).

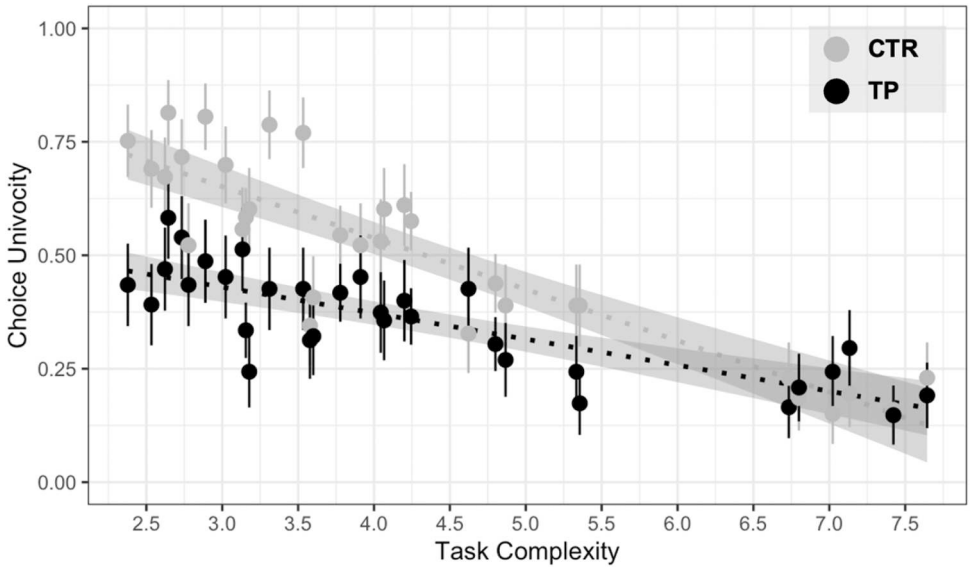


Figure 12. *Choice univocity* vs. *task complexity* for control (CTR, no time pressure) and treatment (TP, time pressure) groups.

participants increasingly relied on the AI's outputs, despite receiving no feedback or performance indicators that could have justified this recalibration of trust. This suggests that mere exposure to the system over time led to a rise in default trust, independent of any rational assessment of its reliability.

These findings highlight a critical risk associated with increased reliance on generally well-performing AI systems, particularly regarding users' diminished ability to identify errors. When an AI system performs well on most tasks or does not provide immediate feedback or performance cues, users are likely to struggle with identifying its occasional mistakes, a challenge that, as we demonstrate here, becomes even more pronounced under time pressure. The resulting reduction in specificity increases the likelihood that AI-generated mistakes go unnoticed. In high-stakes environments such as medical diagnostics, airplane control, or jurisdiction, this failure to detect mistakes could have severe (even fatal) consequences. In contrast, recent research has also shown an opposite effect of algorithm aversion (i.e. the over-correction of trust) based on singular negative outcomes, even though an AI-based system is known to perform superior to a lone human operator (Dietvorst, Simmons, and Massey 2015). In contrast, evidence from clinical decision support shows that, while simple confidence indicators alone have little effect on reliance, richer 'why' explanations can increase trust (and induce over-reliance), while too-sparse explanations or performance cues promote unwarranted self-reliance – implying that feedback design is central to trust calibration (Bussone, Stumpf, and O'Sullivan 2015). Given these risks, it is crucial to implement effective trust calibration mechanisms to mitigate the dangers of unreflected over- or under-reliance on AI. For instance, transparent indicators of an AI system's confidence, periodic human-AI validation checkpoints, and structured verification protocols are essential safeguards, particularly in time-sensitive and high-risk applications where human oversight remains critical.

5.4. Practical implications

With up to 40% of the US population having adopted generative AI for work use (Crane, Green, and Soto 2025), genAI rapidly disrupts how knowledge work is performed within organisations. Our results demonstrate that the degradation of human-AI-team performance under time pressure, task complexity, and AI advice faultiness necessitates careful consideration. Organisations must be careful in their deployment of AI tools, particularly when genAI is involved in the creation of high-stakes decisions or deliverables under conditions that create cognitive load and may trigger heuristic thinking, as is so often the case in modern, hectic work environments. The observed interaction effects suggest that the interface design of productivity-oriented AI solutions should be conceptualised to provide adequate decision support structures while acknowledging the cognitive limitations imposed by time pressure. In high-stakes contexts, AI systems should include risk-tiered safeguards that escalate human oversight and cognition as uncertainty and potential harm increase.

Specifically, AI system designers should consider incorporating transparency mechanisms such as uncertainty cues or confidence indicators that aid users in

judging when outputs might require increased scrutiny or might be incorrect, along with explicit escalation pathways that flag high-risk cases for secondary review. Additionally, adaptive interfaces that recognise time-constrained situations could dynamically simplify task presentation or delay non-urgent actions to reduce user overload. While AI systems may not have direct access to users' external constraints (e.g. looming deadlines), they could infer cognitive strain through proxies such as prompt structure and precision, typing speed, or other interaction patterns indicating stress. In response, AI systems could offer simplified outputs or display a cautionary message, reminding the user of required diligence in dealing with system outputs. Where model uncertainty and/or task risk are high, systems may defer or withhold outputs – or require additional verification steps before release. System inherent trust cues (e.g. anthropomorphic framing and language) should also be carefully calibrated to avoid inducing unwarranted confidence and over-reliance irrespective of time pressure.

In knowledge-intensive environments, where time pressure and heuristic processing are often the norm rather than the exception, the paradoxical relation of task complexity and time pressure hints at an important consideration: The most significant relative performance gains from alleviating time pressure appear to occur in simpler, more routine tasks – tasks that are also the most amenable to automation. In contrast, complex cognitive tasks exhibit comparatively smaller benefits from reduced time pressure. This suggests that organisations aiming to optimise human-AI collaboration should prioritise AI-driven automation for routine tasks while focusing on enhancing cognitive support mechanisms for more complex decision-making processes to mitigate the pitfalls of heuristic decision-making.

The intended role, agency and impact of the AI within the organisation is also to be considered carefully. Recent research from higher education has clearly demonstrated a user preference for '*tutoring*' (i.e. providing step-by-step instructions and solutions) over a '*mentoring*' AI (i.e. providing post-hoc feedback on a human's attempt at a task; Marconi and Cabitza 2025). While the former is perceived as more effective and efficient, it poses an increased risk for overreliance and may prevent reflection, long-term transfer and capability building. Organisations face the challenge of calibrating appropriate trust levels in AI systems, as our findings also suggest that time pressure directly affects performance regardless of users' willingness to rely on AI assistance. This necessitates the development of context-aware systems capable of adapting output confidence and explanation detail based on situational factors (Kocielnik, Amershi, and Bennett 2019). Moreover, training programmes should proactively incorporate scenarios that expose users to high-pressure settings with faulty AI input, helping them understand pitfalls in their specific work domain and build individual metacognitive strategies for error detection.

When negative factors coincide, we observe a significant performance deterioration which might best be referred to as a 'performance cliff' – a phenomenon requiring mitigation through targeted training interventions and workflow restructuring. Furthermore, these findings have implications for regulatory frameworks governing human-AI collaboration, particularly in high-stakes domains where decision quality is paramount. The automation bias literature provides context for our results (Parasuraman and Manzey 2010), but the complex interplay between time constraints and task complexity suggests refinements to existing theoretical models are needed. Organisations must also consider the ethical dimensions of AI deployment, as performance variations under pressure may disproportionately affect different worker populations. The performance metrics employed in evaluating human-AI team effectiveness may need recalibration to account for the nuanced relationship between time pressure and task complexity revealed in this study. Finally, these results call for AI system designers to incorporate adaptivity features that recognise when users operate under conditions likely to impair judgment, potentially offering alternative presentation formats or additional verification mechanisms when algorithm aversion might otherwise occur (Dietvorst, Simmons, and Massey 2015).

As genAI is eagerly adopted by organisations for its capabilities to augment humans, increase efficiency, and speed up the completion of both complex and routine tasks, organisations need to strike a careful balance between embracing the technology for its undoubted benefits and treading carefully to not overrely on the potentially erroneous contribution of the AI colleagues. Especially in potential future settings, where genAI teammates expand human capabilities beyond their ability to effectively supervise it (Sack et al. 2024), unreflected overreliance on genAI may lead to greatly impaired performance and suboptimal outcomes.

5.5. Limitations and directions for future research

Although originally developed for medical AI research, the CLARITY AI research quality checklist offers a structured lens for reporting and evaluating AI studies across core dimensions such as study design, method,

transparency, ethics and rigor (Marconi, Pirovano, and Cabitza 2024). In this spirit, we deem our study to be strongly aligned with the ideal proclaimed by the authors and view it as a useful domain-agnostic tool to evaluate and robustise our work. Nevertheless, as with any research, this study has several limitations. First, the experimental setting, while controlled, can certainly not fully capture the richness of real-world decision-making contexts. While participants were confronted with actual genAI outputs for evaluation, the static presentation of underlying prompts and resulting outputs differed from the conversational user experience typically found in chatbots based on large language models.

The use of a general online sample, though diverse, limits the generalizability of our findings to broader populations and more specific professional settings. Future work may address these issues by incorporating field studies, targeting more specific populations within the domain of knowledge work, and employing either more varied task scenarios or tasks more specific to the workstyle of the population of interest. From a methodological perspective, future research should thus examine the generalizability of these findings across various domains of knowledge work and explore potential moderating factors such as expertise level, domain familiarity, and individual cognitive differences (Bansal et al. 2021).

Longitudinal studies in actual work environments would provide deeper insights into how the dynamics of human-AI interactions evolve; especially as we observed marked shifts in willingness to rely on AI outputs even over the short course of the experiment. Furthermore, expanding the range of task types and including qualitative measures could enrich our understanding of how users adapt their decision strategies in the face of rapidly advancing AI capabilities.

Notes

1. Of course, high-stakes environments do exist, for example when 16,000 Covid-Test results were lost in a UK health office at the height of the pandemic, risking patient lives (The Guardian 2020) or when erroneous calculations by an engineer caused a bridge to collapse, killing 13 and in injuring 145 more (National Transportation Safety Board 2008).
2. This approach means that individual responses may not reflect the full stochastic variability of all possible responses given by LLM-based chatbots to a singular prompt. However, it reflects general user behaviour of generating answers only once and over a set of 36 stimuli, ChatGPT's response variation should be reflected adequately.
3. Setting timed-out tasks to a default value of 5 yields a default *performance* score of 5. It is debatable whether a non-zero *performance* score for timed-out tasks is warranted. As a robustness check, we also computed all following regression models with different imputation methods for *willingness to rely* and *performance*. All effects' directions and general sizes hold up when (1) timed-out tasks are excluded from analysis, or (2) a *willingness to rely* score of 5 is set to reflect indecisiveness, but *performance* is set to 0 for timed-out tasks.
4. All effects described below persist and remain significant if *choice univocity* is relaxed to also include a stated willingness of 1,2, 8 or 9 for almost univocal choices. Naturally, the share of univocal choices increases with this relaxed definition, but direction, intensity, and difference of effects between groups remain constant. Appendix E outlines the share of univocal choices for relaxed definitions of *choice univocity*.

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Author contributions

CRediT: **Lukas Hermanns:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Visualization, Writing – original draft, Writing – review & editing; **Timm Teubner:** Conceptualization, Formal analysis, Methodology, Supervision, Writing – review & editing.

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References

Ackerman, R., and V. A. Thompson. 2017. "Meta-reasoning: Monitoring and Control of Thinking and Reasoning." *Trends in Cognitive Sciences* 21 (8): 607–617. https://doi.org/10.1016/j.tics.2017.05.004.

Banovic, N., Z. Yang, A. Ramesh, and A. Liu. 2023. "Being Trustworthy Is Not Enough: How Untrustworthy Artificial Intelligence (AI) Can Deceive the end-Users

and Gain Their Trust." *Proceedings of the ACM on Human-Computer Interaction* 7:1–17. https://doi.org/10.1145/3579460.

Bansal, G., B. Nushi, E. Kamar, E. Horvitz, and D. S. Weld. 2021. "Is the Most Accurate AI the Best Teammate? Optimizing AI for Teamwork." *Proceedings of the AAAI Conference on Artificial Intelligence*. 2021, May. https://doi.org/10.1609/aaai.v35i13.17359.

Bedny, G. Z., W. Karwowski, and I. S. Bedny. 2012. "Complexity Evaluation of Computer-Based Tasks." *International Journal of Human-Computer Interaction* 28 (4): 236–257. https://doi.org/10.1080/10447318.2011.581895.

Benbya, H., S. Pachidi, and S. L. Jarvenpaa. 2021. "Special Issue Editorial: Artificial Intelligence in Organizations: Implications for Information Systems Research." *Journal of the Association for Information Systems* 22 (2): 281–303. https://doi.org/10.17705/1jais.00662.

Benbya, H., F. Strich, and T. Tamm. 2024. "Navigating Generative Artificial Intelligence Promises and Perils for Knowledge and Creative Work." *Journal of the Association for Information Systems* 25 (1): 23–36. https://doi.org/10.17705/1jais.00861.

Borji, A. 2023. *A Categorical Archive of ChatGPT Failures*. http://arxiv.org/abs/2302.03494.

Brynjolfsson, E., and T. Mitchell. 2017. "What Can Machine Learning Do? Workforce Implications." *Science* 358 (6370): 1530–1534. https://doi.org/10.1126/science.aap8062.

Bueno-Vesga, J. A., X. Xu, and H. He. 2021. "The Effects of Cognitive Load on Engagement in a Virtual Reality Learning Environment." *Proceedings of the 2021 IEEE Virtual Reality and 3D User Interfaces (VR)*. Lisboa, Portugal. 2021. https://doi.org/10.1109/VR50410.2021.00090.

Bussone, A., S. Stumpf, and D. O'Sullivan. 2015. "The Role of Explanations on Trust and Reliance in Clinical Decision Support Systems." *2015 International Conference on Healthcare Informatics* 14:160–169. https://doi.org/10.1109/ICHI.2015.26.

Cacioppo, J. T., R. E. Petty, C. F. Kao, and R. Rodriguez. 1986. "Central and Peripheral Routes to Persuasion: An Individual Difference Perspective." *Journal of Personality and Social Psychology* 51 (5): 1032–1043. https://doi.org/10.1037/0022-3514.51.5.1032.

Cao, S., C. Gomez, and C.-M. Huang. 2023. "How Time Pressure in Different Phases of Decision-Making Influences Human-AI Collaboration." *Proceedings of the ACM on Human-Computer Interaction* 7:1–26. https://doi.org/10.1145/3610068.

Chaiken, S. 1980. "Heuristic versus Systematic Information Processing and the use of Source versus Message Cues in Persuasion." *Journal of Personality and Social Psychology* 39 (5): 752–766. https://doi.org/10.1037/0022-3514.39.5.752.

Chanda, A. K. 2024. "Human Judgement in Artificial Intelligence for Business Decision-Making: An Empirical Study." *International Journal of Innovation Management* 28 (1+2): 1–29. https://doi.org/10.1142/S136391962450004X.

Crane, L., M. Green, and P. Soto. 2025. *Measuring AI Uptake in the Workplace*. United States Federal Reserve. https://www.federalreserve.gov/econres/notes/feds-notes/measuring-ai-uptake-in-the-workplace-20240205.html.

Dăniloaia, D.-F., and E. Turturean. 2024. "Knowledge Workers and the Rise of Artificial Intelligence: Navigating new Challenges." *SEA - Practical Application of Science* 12 (35): 111–121. https://doi.org/10.70147/s35111121.

Daronnat, S., L. Azzopardi, M. Halvey, and M. Dubiel. 2021. "Inferring Trust from Users' Behaviours; Agents' Predictability Positively Affects Trust, Task Performance and Cognitive Load in Human-Agent Real-Time Collaboration." *Frontiers in Robotics and AI* 8:1–14. https://doi.org/10.3389/frobt.2021.642201.

Dell'Acqua, F., E. McFowland, E. R. Mollick, H. Lifshitz-Assaf, K. Kellogg, S. Rajendran, L. Krayer, F. Candelon, and K. R. Lakhani. 2023. "Navigating the Jagged Technological Frontier: Field Experimental Evidence of the Effects of AI on Knowledge Worker Productivity and Quality." *SSRN Electronic Journal* 4:24–013. https://doi.org/10.2139/ssrn.4573321.

Dietvorst, B. J., J. P. Simmons, and C. Massey. 2015. "Algorithm Aversion: People Erroneously Avoid Algorithms after Seeing Them err." *Journal of Experimental Psychology: General* 144 (1): 114–126. https://doi.org/10.1037/xge0000033.

Dohmen, T., A. Falk, D. Huffman, U. Sunde, J. Schupp, and G. G. Wagner. 2011. "Individual Risk Attitudes: Measurement, Determinants, and Behavioral Consequences." *Journal of the European Economic Association* 9 (3): 522–550. https://doi.org/10.1111/j.1542-4774.2011.01015.x.

Douglas, B. D., P. J. Ewell, and M. Brauer. 2023. "Data Quality in Online Human-Subjects Research: Comparisons between MTurk, Prolific, CloudResearch, Qualtrics, and SONA." *PLoS One* 18 (3): e0279720. https://doi.org/10.1371/journal.pone.0279720.

Dror, I. E., B. Basola, and J. R. Busemeyer. 1999. "Decision Making under Time Pressure: An Independent Test of Sequential Sampling Models." *Memory & Cognition* 27 (4): 713–725. https://doi.org/10.3758/BF03211564.

Eurostat. 2020. *One in Three EU workers Experience Time Pressure - Products Eurostat News - Eurostat*. Eurostat. https://ec.europa.eu/eurostat/web/products-eurostat-news/-/ddn-20201006-2.

Evans, J., B. T. St, and K. E. Stanovich. 2013. "Dual-process Theories of Higher Cognition." *Perspectives on Psychological Science* 8 (3): 223–241. https://doi.org/10.1177/1745691612460685.

Flavell, J. H. 1979. "Metacognition and Cognitive Monitoring: A new Area of Cognitive–Developmental Inquiry." *American Psychologist* 34 (10): 906–911. https://doi.org/10.1037/0003-066X.34.10.906.

Gigerenzer, G., and W. Gaissmaier. 2011. "Heuristic Decision Making." *Annual Review of Psychology* 62 (1): 451–482. https://doi.org/10.1146/annurev-psych-120709-145346.

Gruetzemacher, R., and J. Whittlestone. 2022. "The Transformative Potential of Artificial Intelligence." *Futures* 135 (102884): 102884–11. https://doi.org/10.1016/j.futures.2021.102884.

Haefner, N., J. Wincent, V. Parida, and O. Gassmann. 2021. "Artificial Intelligence and Innovation Management: A Review, Framework, and Research Agenda." *Technological Forecasting and Social Change* 162 (120392): 120392–10. https://doi.org/10.1016/j.techfore.2020.120392.

Hancock, P. A., D. R. Billings, K. E. Schaefer, J. Y. C. Chen, E. J. de Visser, and R. Parasuraman. 2011. "A Meta-analysis of Factors Affecting Trust in Human-Robot Interaction." *Human Factors: The Journal of the Human Factors and*

Ergonomics Society 53 (5): 517–527. https://doi.org/10.1177/0018720811417254.

Hurtienne, J., U. Stilijanow, and G. Junghanns. 2014. "Time and Work Pressure in Today's Working World." In *The Impact of ICT on Quality of Working Life*, edited by C. Korunka and P. Hoonakker, 63–85. Dordrecht: Springer.

Hwang, M. I. 1994. "Decision Making under Time Pressure: A Model for Information Systems Research." *Information & Management* 27 (4): 197–203. https://doi.org/10.1016/0378-7206(94)90048-5.

Jarke-Neuert, J., and J. Lohse. 2022. "I'm in a Hurry, I Don't Want to Know! Strategic Ignorance under Time Pressure." *Journal of Experimental Psychology: General* 151 (11): 2833–2845. https://doi.org/10.1037/xge0001222.

Jarrahi, M. H. 2018. "Artificial Intelligence and the Future of Work: Human-AI Symbiosis in Organizational Decision Making." *Business Horizons* 61 (4): 577–586. https://doi.org/10.1016/j.bushor.2018.03.007.

Kahneman, D. 2011. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York.

Kahneman, D., and G. Klein. 2009. "Conditions for Intuitive Expertise: A Failure to Disagree." *American Psychologist* 64 (6): 515–526. https://doi.org/10.1037/a0016755.

Kerstholt, J. H. 1994. "The Effect of Time Pressure on Decision-Making Behaviour in a Dynamic Task Environment." *Acta Psychologica* 86 (1): 89–104. https://doi.org/10.1016/0001-6918(94)90013-2.

Kleinmuntz, D. N. 1985. "Cognitive Heuristics and Feedback in a Dynamic Decision Environment." *Management Science* 31 (6): 680–702. https://doi.org/10.1287/mnsc.31.6.680.

Kocielnik, R., S. Amershi, and P. N. Bennett. 2019. "Will You Accept an Imperfect AI?" *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* 1:1–14. https://doi.org/10.1145/3290605.3300641.

Koć-Januchta, M. M., K. J. Schönborn, C. Roehrig, V. K. Chaudhri, L. A. E. Tibell, and H. C. Heller. 2022. "Connecting Concepts Helps put Main Ideas Together: Cognitive Load and Usability in Learning Biology with an AI-Enriched Textbook." *International Journal of Educational Technology in Higher Education* 19 (1): 1–22. https://doi.org/10.1186/s41239-021-00317-3.

Krügel, S., A. Ostermaier, and M. Uhl. 2022. "Zombies in the Loop? Humans Trust Untrustworthy AI-Advisors for Ethical Decisions." *Philosophy & Technology* 35 (1): 1–37. https://doi.org/10.1007/s13347-022-00511-9.

Lai, Y., A. Kankanhalli, and D. C. Ong. 2021. "Human-AI Collaboration in Healthcare: A Review and Research Agenda." *Proceedings of the 2021 HICSS* 18:390–399. https://hdl.handle.net/10125/70657.

Leilabadi, S. H., and S. Schmidt. 2019. "In-depth Analysis of Autonomous Vehicle Collisions in California." *Proceedings of the 2019 IEEE Intelligent Transportation Systems Conference (ITSC)*. Auckland, New Zealand. https://doi.org/10.1109/ITSC.2019.8916775.

Lenat, D., and G. Marcus. 2023. *Getting from Generative AI to Trustworthy AI: What LLMs Might Learn from Cyc*. http://arxiv.org/abs/2308.04445.

Lerner, J. S., and D. Keltner. 2001. "Fear, Anger, and Risk." *Journal of Personality and Social Psychology* 81 (1): 146–159. https://doi.org/10.1037/0022-3514.81.1.146.

Lieder, F., and T. L. Griffiths. 2020. "Resource-rational Analysis: Understanding Human Cognition as the Optimal use of Limited Computational Resources." *Behavioral and Brain Sciences* 43:e1. https://doi.org/10.1017/S0140525X1900061X.

Macrae, C. 2019. "Governing the Safety of Artificial Intelligence in Healthcare." *BMJ Quality & Safety* 28 (6): 495–498. https://doi.org/10.1136/bmjqs-2019-009484.

Madan, C. R., M. L. Spetch, and E. A. Ludvig. 2015. "Rapid Makes Risky: Time Pressure Increases Risk Seeking in Decisions from Experience." *Journal of Cognitive Psychology* 27 (8): 921–928. https://doi.org/10.1080/20445911.2015.1055274.

Marconi, L., and F. Cabitza. 2025. *Tutor or Mentor - A Comparative Usability Study of AI Roles in Higher Education* (pp. 201–208). https://doi.org/10.1007/978-3-031-99267-4_25

Marconi, L., E. Pirovano, and F. Cabitza. 2024. "CLARITY AI: A Comprehensive Checklist Integrating Established Frameworks for Enhanced Research Quality in Medical AI Studies." *CEUR Workshop Proceedings* 3880:1–14.

Marewski, J. N., and G. Gigerenzer. 2012. "Heuristic Decision Making in Medicine." *Dialogues in Clinical Neuroscience* 14 (1): 77–89. https://doi.org/10.31887/DCNS.2012.14.1/jmarewski.

Merritt, S. M., H. Heimbaugh, J. Lachapell, and D. Lee. 2013. "I Trust It, but I Don't Know why: Effects of Implicit Attitudes toward Automation on Trust in an Automated System." *Human Factors* 55 (3): 520–534. https://doi.org/10.1177/0018720812465081.

Mulukuntla, S. 2022. "Generative AI - Benefits, Limitations, Potential Risks and Challenges in Healthcare Industry." *EPH - International Journal of Medical and Health Science* 8 (4): 1–9. https://doi.org/10.53555/eijmhs.v8i4.211.

National Transportation Safety Board. 2008. *Collapse of I-35W Highway Bridge*. National Transportation Safety Board. https://www.ntsb.gov/investigations/Pages/HWY07MH024.aspx.

Ninaus, K., S. Diehl, R. Terlutter, K. Chan, and A. Huang. 2015. "Benefits and Stressors – Perceived Effects of ICT use on Employee Health and Work Stress: An Exploratory Study from Austria and Hong Kong." *International Journal of Qualitative Studies on Health and Well-Being* 10 (1): 28838. https://doi.org/10.3402/qhw.v10.28838.

Nov, O., N. Singh, and D. Mann. 2023. "Putting ChatGPT's Medical Advice to the (Turing) Test: Survey Study." *JMIR Medical Education* 9 (e46939): e46939. https://doi.org/10.2196/46939.

Palani, S., and G. Ramos. 2024. "Evolving Roles and Workflows of Creative Practitioners in the age of Generative AI." *Proceedings of the 16th Conference on Creativity & Cognition*.Chicago, USA. 2024, June. https://doi.org/10.1145/3635636.3656190.

Parasuraman, R., and D. H. Manzey. 2010. "Complacency and Bias in Human use of Automation: An Attentional Integration." *Human Factors: The Journal of the Human Factors and Ergonomics Society* 52 (3): 381–410. https://doi.org/10.1177/0018720810376055.

Passi, S., and M. Vorvoreanu. 2022. "Overreliance on AI: Literature Review." *Microsoft Research* 1:1–24. https://www.microsoft.com/en-Us/research/uploads/prod/2022/06/Aether-Overreliance-on-AI-Review-Final-6.21.22.pdf.

Payne, J. W., J. R. Bettman, and E. J. Johnson. 1993. *The Adaptive Decision Maker*. Cambridge: Cambridge University Press.

Peer, E., L. Brandimarte, S. Samat, and A. Acquisti. 2017. "Beyond the Turk: Alternative Platforms for Crowdsourcing Behavioral Research." *Journal of Experimental Social Psychology* 70:153–163. https://doi.org/10.1016/j.jesp.2017.01.006.

Peer, E., D. Rothschild, A. Gordon, Z. Evernden, and E. Damer. 2021. "Data Quality of Platforms and Panels for Online Behavioral Research." *Behavior Research Methods* 54 (4): 1643–1662. https://doi.org/10.3758/s13428-021-01694-3.

Rahwan, I., M. Cebrian, N. Obradovich, J. Bongard, J.-F. Bonnefon, C. Breazeal, J. W. Crandall, et al. 2019. "Machine Behaviour." *Nature* 568 (7753): 477–486. https://doi.org/10.1038/s41586-019-1138-y.

Ranyard, R., Wr. Crozier, and O. Svenson. 2002. *Decision Making*. Milton Park: Routledge.

Rice, S., and D. Trafimow. 2012. "Time Pressure Heuristics Can Improve Performance due to Increased Consistency." *The Journal of General Psychology* 139 (4): 273–288. https://doi.org/10.1080/00221309.2012.705187.

Ritala, P., M. Ruokonen, and L. Ramaul. 2024. "Transforming Boundaries: How Does ChatGPT Change Knowledge Work?" *Journal of Business Strategy* 45 (3): 214–220. https://doi.org/10.1108/JBS-05-2023-0094.

Rodenburg, D., P. Hungler, S. A. Etemad, D. Howes, A. Szulewski, and J. Mclellan. 2018. "Dynamically Adaptive Simulation Based on Expertise and Cognitive Load." *Proceedings of 2018 IEEE GEM* 0:1–6. https://doi.org/10.1109/GEM.2018.8587618.

Russo, D. 2022. "Recruiting Software Engineers on Prolific." *In RoPES 2022 Proceedings* 0:1–2. http://arxiv.org/abs/2203.14695.

Sabherwal, R., and V. Grover. 2024. "The Societal Impacts of Generative Artificial Intelligence: A Balanced Perspective." *Journal of the Association for Information Systems* 25 (1): 13–22. https://doi.org/10.17705/1jais.00860.

Sachin, P. K., and A. Schecter. 2024. "Advice Utilization in Combined Human-Algorithm Decision-Making: An Analysis of Preferences and Behaviors." *Journal of the Association for Information Systems* 25 (6): 1439–1465. https://doi.org/10.17705/1jais.00896.

Sack, D., L. Krayer, E. Wiles, M. Abbadi, U. Awasthi, R. Kennedy, C. Arnolds, and F. Candelon. 2024. *GenAI Doesn't Just Increase Productivity. It Expands Capabilities*. https://www.bcg.com/publications/2024/gen-ai-increases-productivity-and-expands-capabilities.

Schöffer, J., J. Jakubik, M. Vössing, N. Kühl, and G. Satzger. 2025. "AI Reliance and Decision Quality: Fundamentals, Interdependence, and the Effects of Interventions." *Journal of Artificial Intelligence Research* 82:471–501. https://doi.org/10.1613/jair.1.15873.

Securities and Exchange Commission. 2013. *In the matter of New York Stock Exchange LLC, and NYSE Euronext*. http://www.sec.gov/litigation/admin/2012/34-67857.pdf.

Securities and Exchange Commission, & Commodity Futures Trading Commission. 2010. *Findings regarding the market events of May 6*, 2010. www.cftc.gov.

Seeber, I., E. Bittner, R. O. Briggs, T. de Vreede, G.-J. de Vreede, A. Elkins, R. Maier, et al. 2020. "Machines as Teammates: A Research Agenda on AI in Team Collaboration." *Information & Management* 57 (2): 103174. https://doi.org/10.1016/j.im.2019.103174.

Shin, D., B. Zaid, and M. Ibahrine. 2020. "Algorithm Appreciation: Algorithmic Performance, Developmental Processes, and User Interactions." *Proceedings of the 2020 CCCI* 0:1–5. https://doi.org/10.1109/CCCI49893.2020.9256470.

Simon, H. A. 1955. "A Behavioral Model of Rational Choice." *The Quarterly Journal of Economics* 69 (1): 99–118. https://doi.org/10.2307/1884852.

Solaiman, I., Z. Talat, W. Agnew, L. Ahmad, D. Baker, S. L. Blodgett, C. Chen, et al. 2023. *Evaluating the Social Impact of Generative AI Systems in Systems and Society*. http://arxiv.org/abs/2306.05949.

Spiliopoulos, L., A. Ortmann, and L. Zhang. 2018. "Complexity, Attention, and Choice in Games under Time Constraints: A Process Analysis." *Journal of Experimental Psychology: Learning, Memory, and Cognition* 44 (10): 1609–1640. https://doi.org/10.1037/xlm0000535.

Starcke, K., and M. Brand. 2012. "Decision Making under Stress: A Selective Review." *Neuroscience & Biobehavioral Reviews* 36 (4): 1228–1248. https://doi.org/10.1016/j.neubiorev.2012.02.003.

Sweller, J. 1988. "Cognitive Load during Problem Solving: Effects on Learning." *Cognitive Science* 12 (2): 257–285. https://doi.org/10.1016/0364-0213(88)90023-7.

Sweller, J. 2011. "Cognitive Load Theory." In *The Psychology of Learning and Motivation*, edited by Jose. P. Mestre and B. H. Ross, 37–76. Cambridge, MA: Academic Press.

Tang, N., C. Yang, J. Fan, L. Cao, Y. Luo, and A. Halevy. 2023. *VerifAI: Verified generative AI*. http://arxiv.org/abs/2307.02796.

The Guardian. 2020. *Covid: How Excel May Have Caused Loss of 16,000 Test Results in England*. The Guardian. https://www.theguardian.com/politics/2020/oct/05/How-excel-May-Have-caused-loss-of-16000-covid-tests-in-england.

Watts, S. 2015. "Application of Dual-Process Theory to Information Systems: Current and Future Research Directions." *Foundations and Trends in Information Systems* 1 (2): 69–162. https://doi.org/10.1561/2900000004.

Weidinger, L., M. Rauh, N. Marchal, A. Manzini, L. A. Hendricks, J. Mateos-Garcia, S. Bergman, et al. 2023. *Sociotechnical Safety Evaluation of Generative AI Systems*. http://arxiv.org/abs/2310.11986.

Weisz, J. D., M. Muller, J. He, and S. Houde. 2023. *Toward General Design Principles for Generative AI Applications*. http://arxiv.org/abs/2301.05578.

Wienrich, C., and M. E. Latoschik. 2021. "eXtended Artificial Intelligence: New Prospects of Human-AI Interaction Research." *Frontiers in Virtual Reality* 2 (686783): 1–19. https://doi.org/10.3389/frvir.2021.686783.

Wilson, H. J., and P. R. Daugherty. 2018. "Collaborative Intelligence: Humans and AI Are Joining Forces." *Harvard Business Review* 4 (96): 114–123.

Wood, R. E. 1986. "Task Complexity: Definition of the Construct." *Organizational Behavior and Human Decision Processes* 37 (1): 60–82. https://doi.org/10.1016/0749-5978(86)90044-0.

Young, D. L., A. S. Goodie, D. B. Hall, and E. Wu. 2012. "Decision Making under Time Pressure, Modeled in a Prospect Theory Framework." *Organizational Behavior and Human Decision Processes* 118 (2): 179–188. https://doi.org/10.1016/j.obhdp.2012.03.005.